

Control of industrial robots

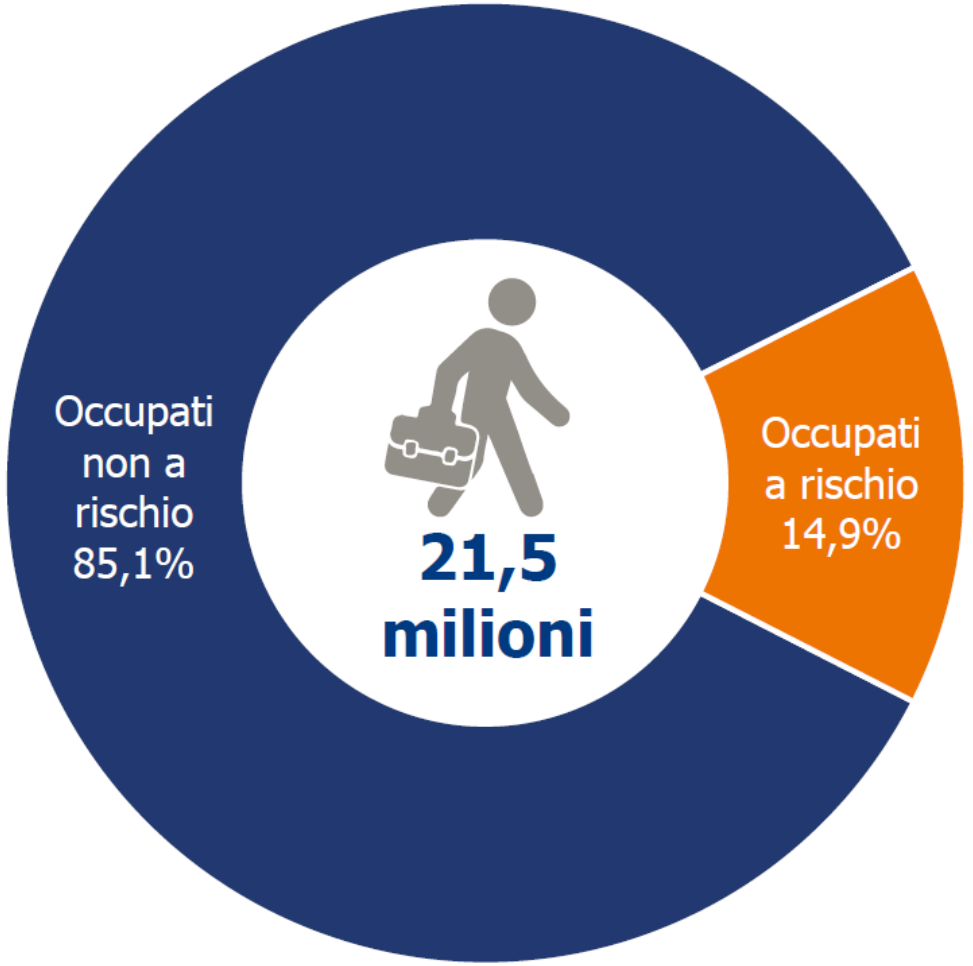
Collaborative robotics

Prof. Paolo Rocco (paolo.rocco@polimi.it)

Politecnico di Milano, Dipartimento di Elettronica, Informazione e Bioingegneria

Automation: a risk or an opportunity?



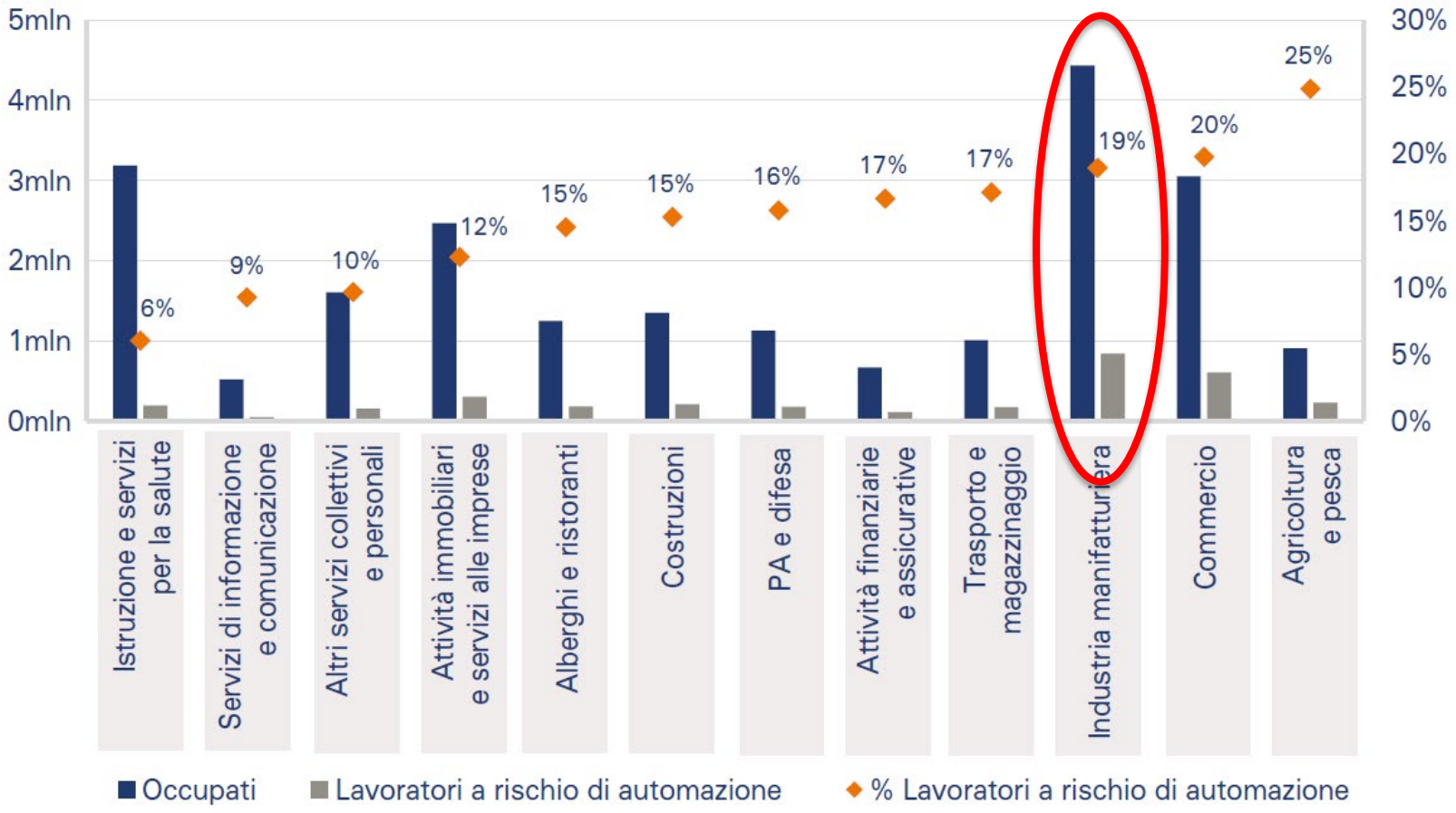


Fonte: The European House - Ambrosetti

Automation: a risk or an opportunity?



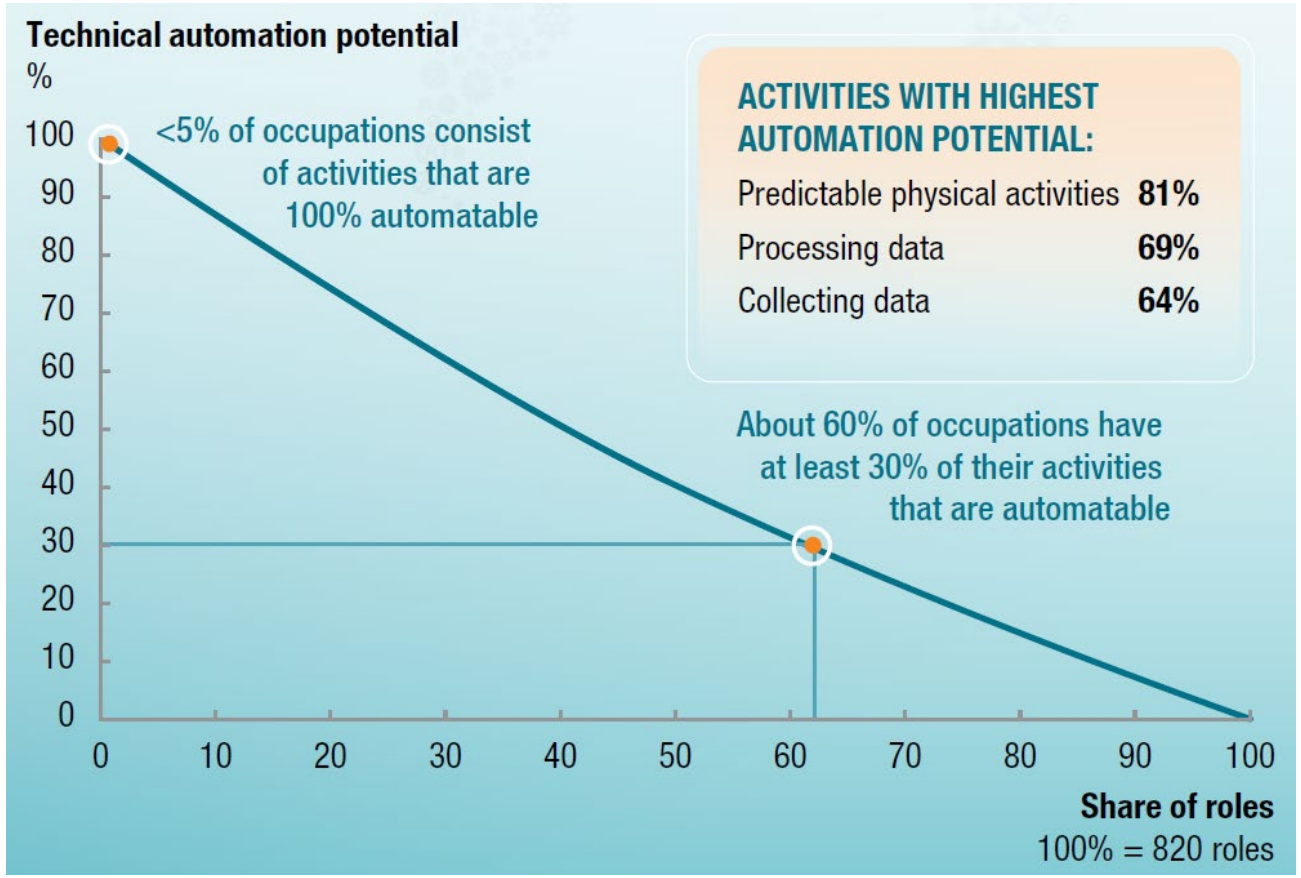
Fonte: The European House - Ambrosetti



Potential of automation of the activities



Source: McKinsey Global Institute



Automation, a risk or an opportunity?



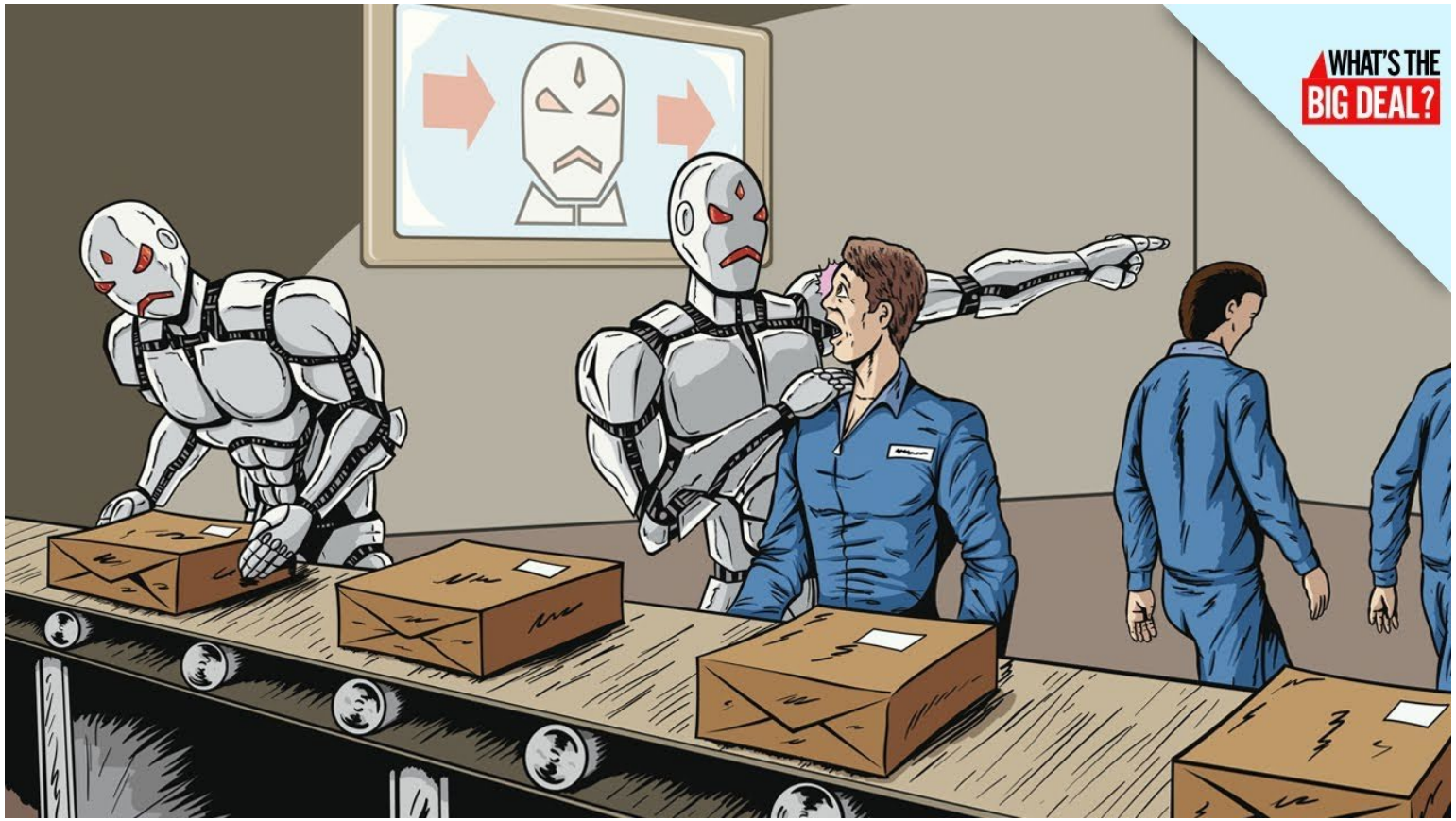
Source: McKinsey Global Institute

Time spent on activities that can be automated by adapting currently demonstrated technology
%

	Manage[1]	Expertise[2]	Interface[3]	Unpredict-able physical[4]	Collect data	Process data	Predictable physical[5]
Time spent in all US occupations %	7	14	16	12	17	16	18
	9	18	20	26	64	69	81
Total wages in US, 2014 $ billion	596	1,190	896	504	1,030	931	766

Most susceptible activities 51% of total employment $2.7 trillion in wages

Will workers disappear from the factories?



… not necessarily



Fonte: Universal Robots

Collaborative robotics

Collaborative robotics is a new paradigm in industrial robotics, where humans and robots share the same environment and collaborate at the same tasks.

New collaborative robots (cobots) are now entering the market and all the major robot manufacturers have their collaborative solution.

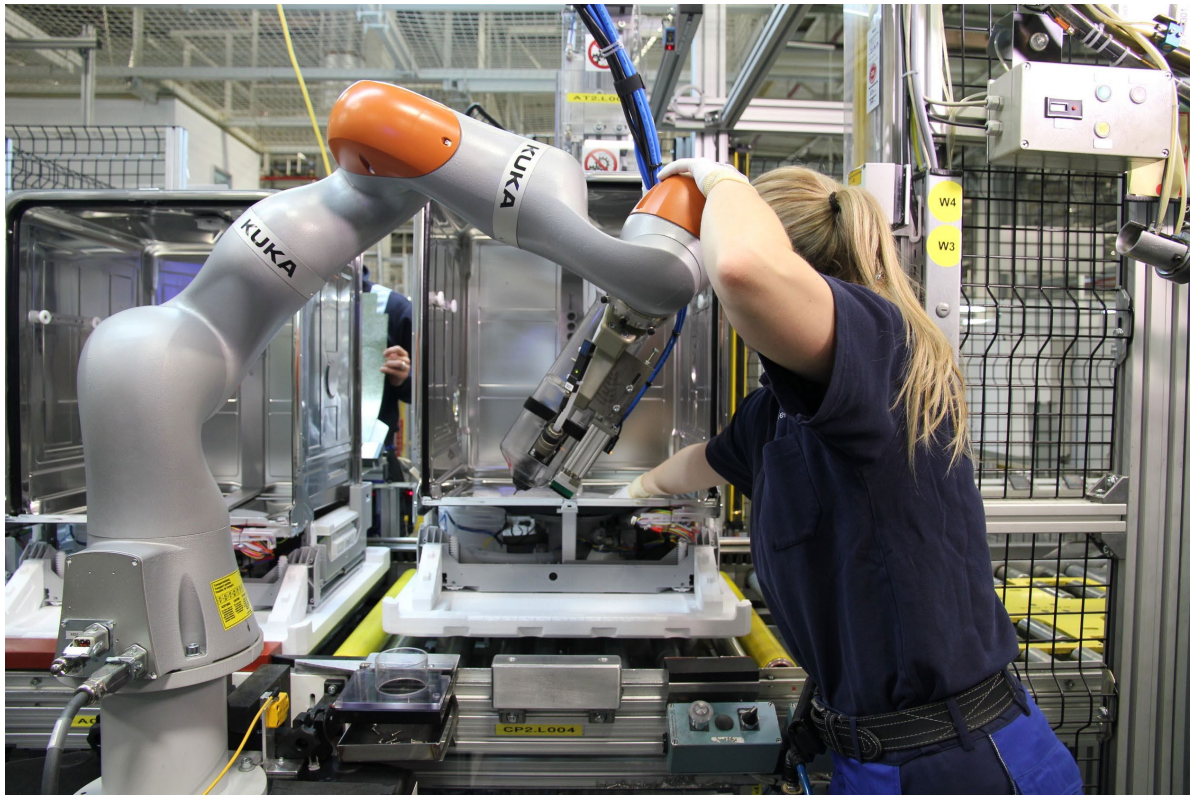
Though still a niche in the big market of industrial robotics, collaborative robotics is growing fast and is expected to be a breakthrough in the coming years.



Source: Universal Robots

Collaborative robotics (1/3)

- Humans and robots collaborating at the same task
- Protective fences are not needed
- Particularly interesting for SMEs (reduced cost, reduced foot print)



Source: KUKA

Collaborative robotics (2/3)

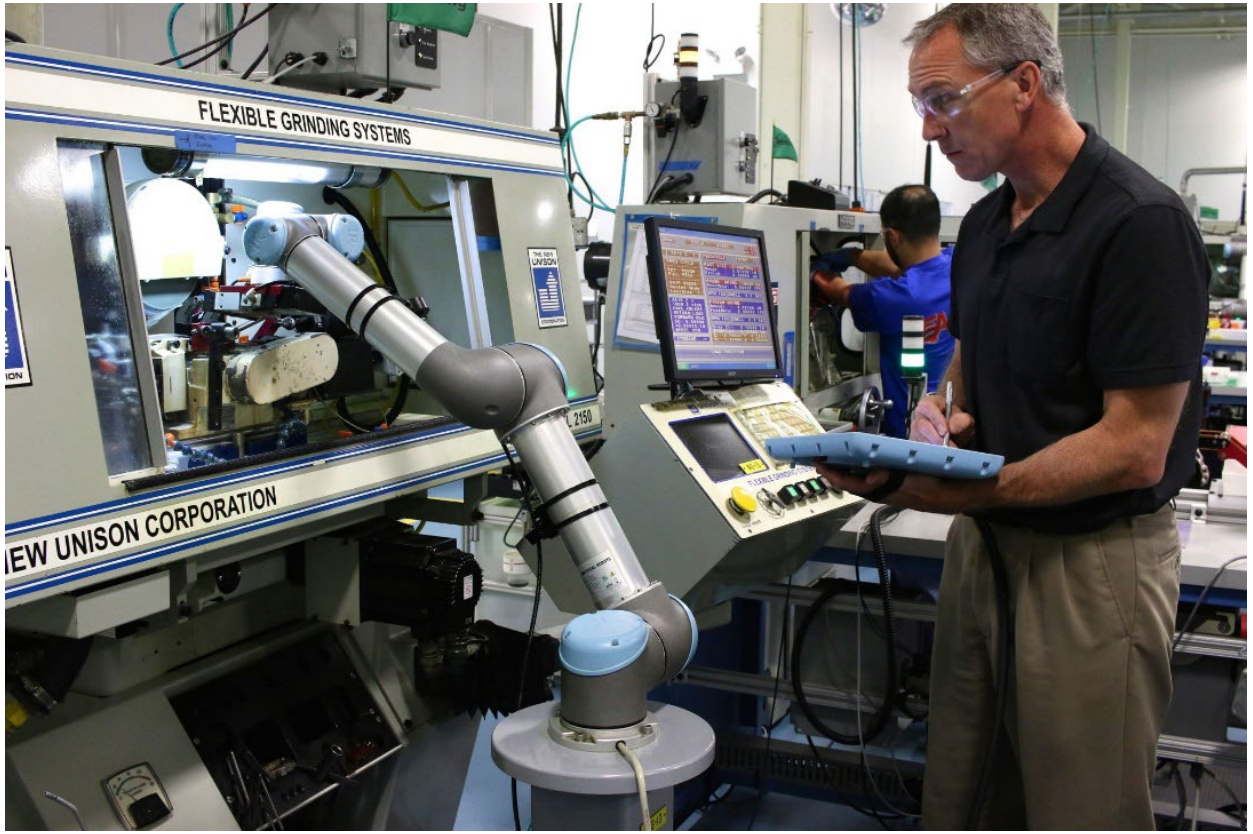
- Redundant, dual arm manipulators, characterized by low inertia and low payload (reduced risk when impact)
- Still good precision available
- Good potential for assembly of electronic parts



Source: ABB

Collaborative robotics (3/3)

- New programming interfaces
- Decrease of the deployment time
- High potential for SMEs

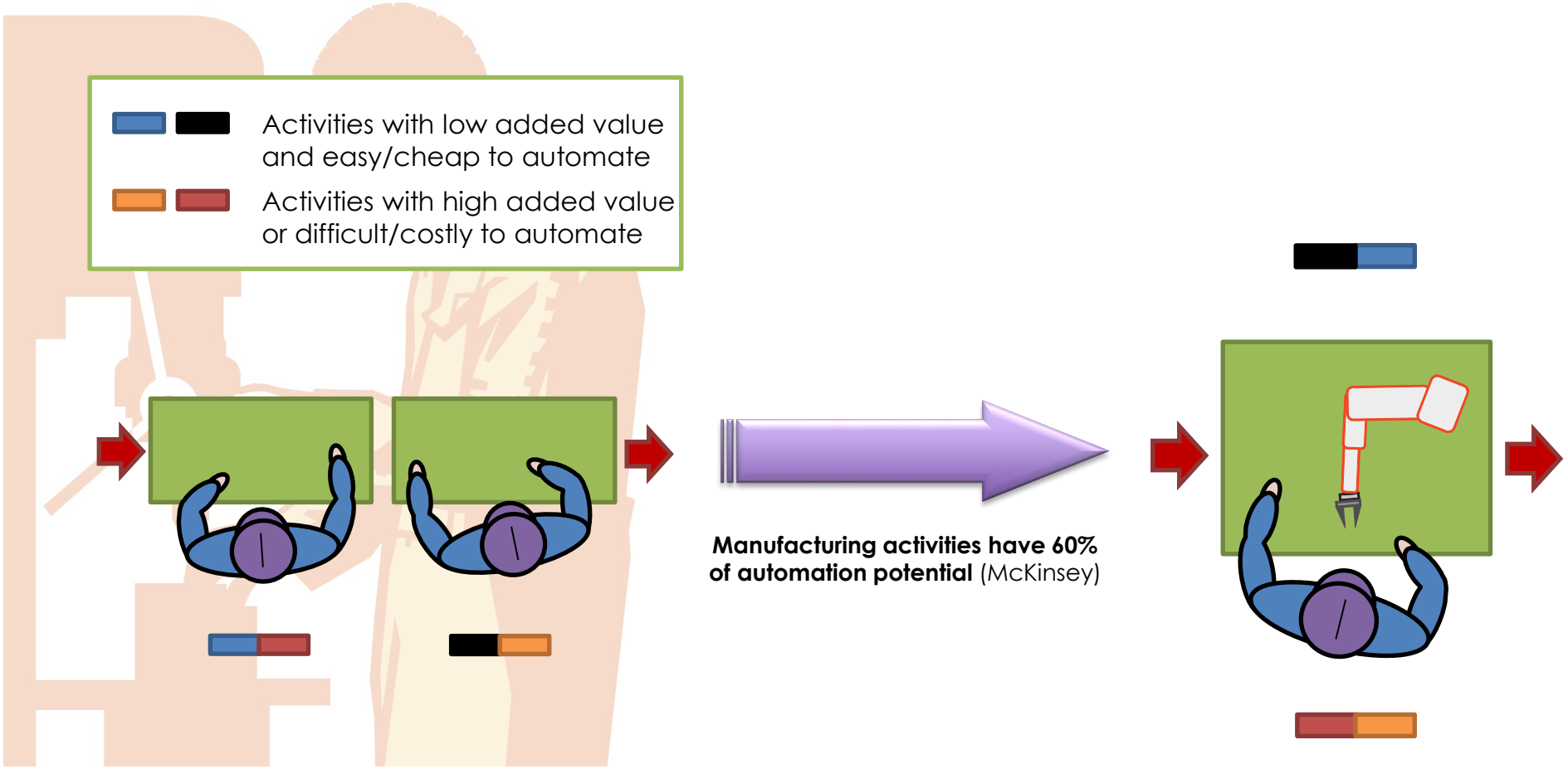


Source: Universal Robots

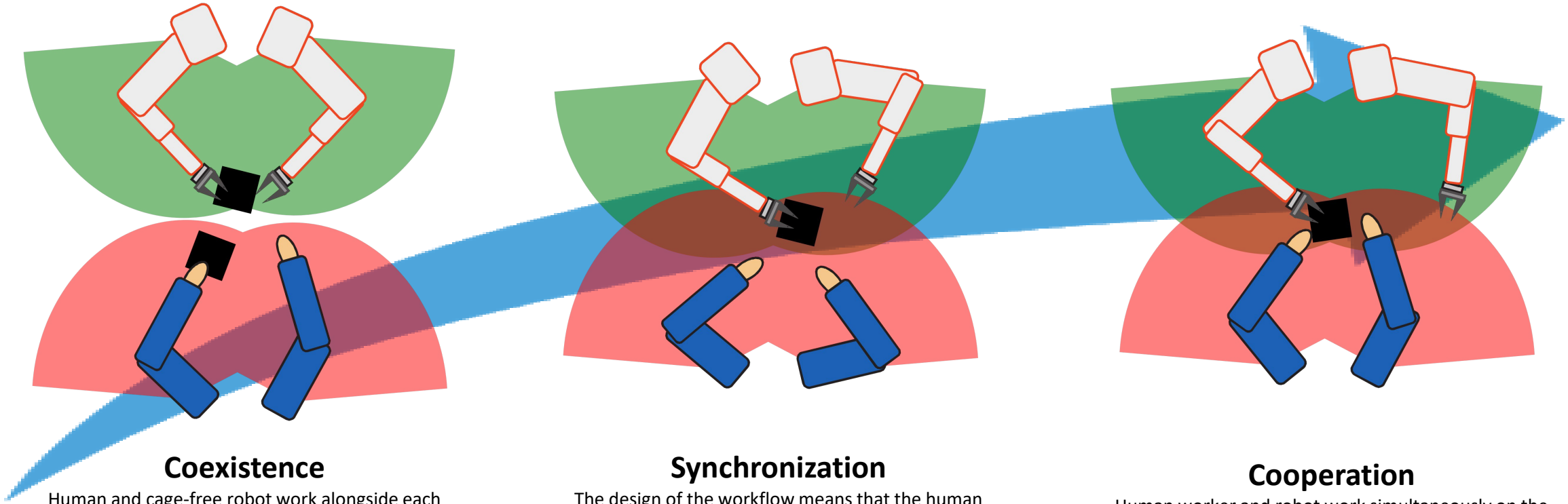
Collaborative assembly



Opportunities



Interaction modes



Coexistence

Human and cage-free robot work alongside each other but do not share a workspace. Humans can work away, but may need to *sporadically* access the workspace of the robot (e.g. to load/unload parts).

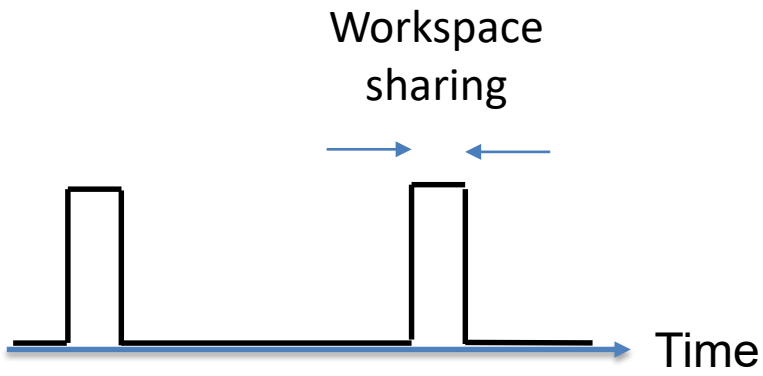
Synchronization

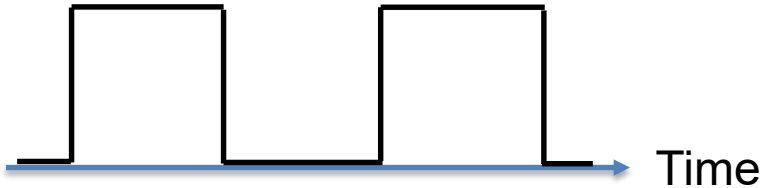
The design of the workflow means that the human worker and the robot share a workspace but that only one of the interaction partners is actually present in the workspace at any one time, or they do not work simultaneously on the same product or component.

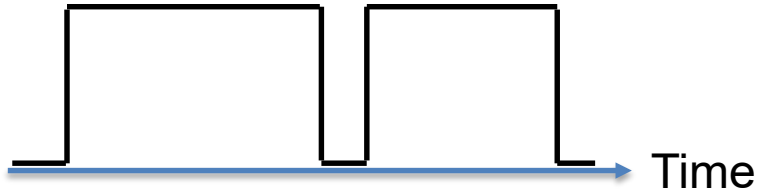
Cooperation

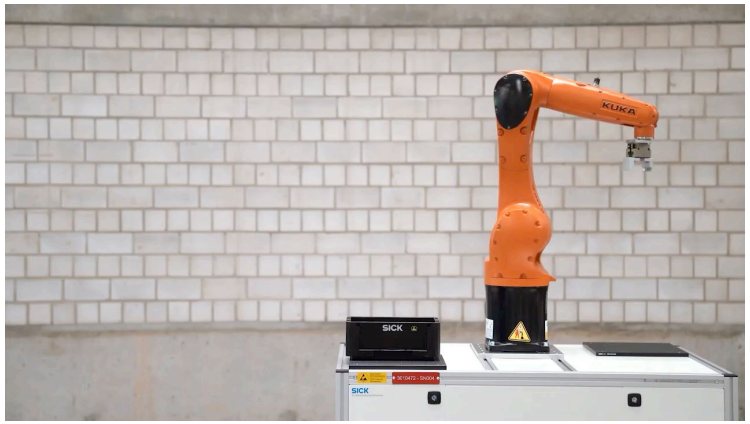
Human worker and robot work simultaneously on the same product or component.

Workspace sharing









Sporadic



Frequent



Persistent

A bit of history of collaborative robotics

2000 | 2000 | 2002 | 2004 | 2006 | 2008 | 2010 | 2012 | 2014 | 2016 | 2017

1960 - 2000
Industrial robots are separated from humans; they requires significant investments and programming skills

2000
The world cobot enters the dictionary
cobot: noun. A collaborative robot designed to help workers on the job instead of replacing them. Source: Wall Street Journal, Jan 2000

2005
Universal Robots is founded in Denmark to develop an affordable, low-cost collaborative robot
UR5 was launched in 2008 and certified by TUV in 2014

2012
Rethink Robotics is founded and launches Baxter

2013
KUKA launches LBR iiwa

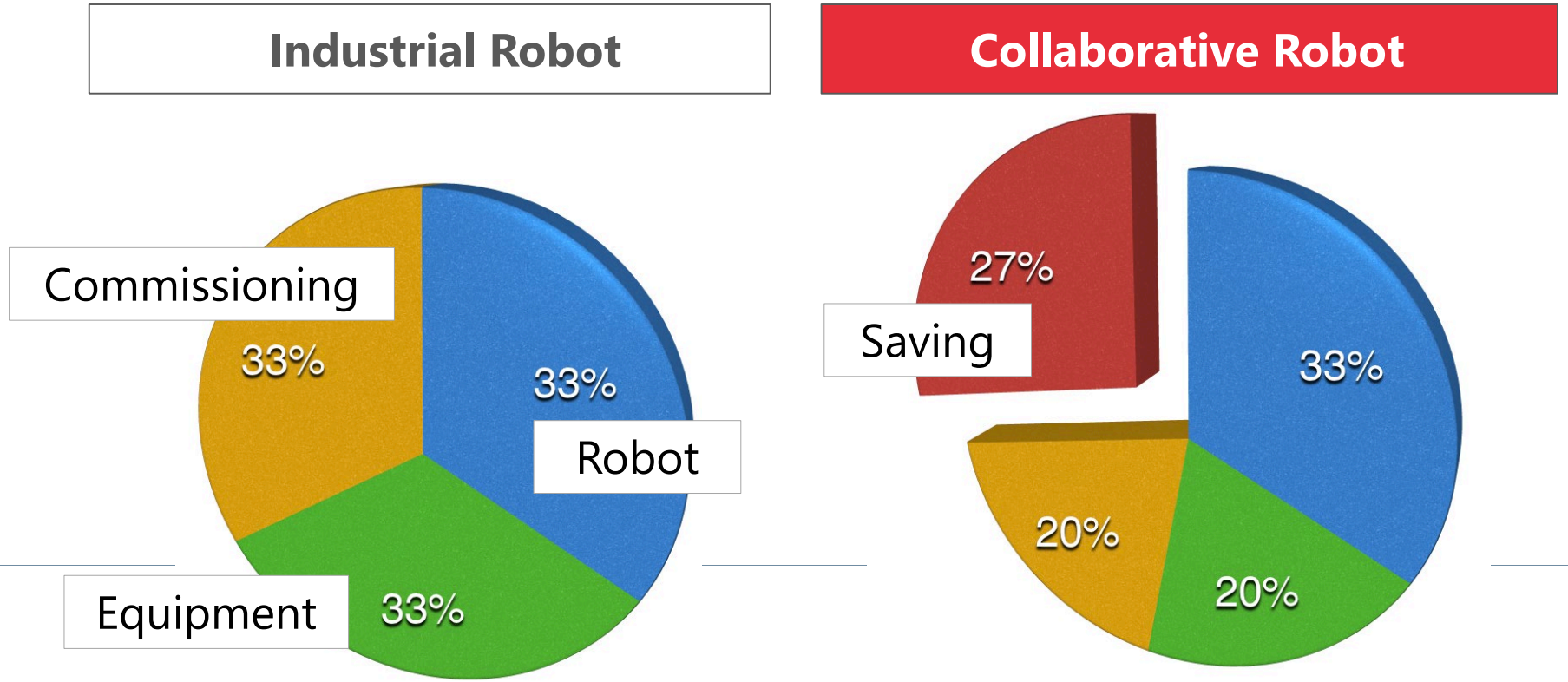
2015
ABB launches YuMi and acquires Gomtec

2016
ISO sets a new standard for cobots

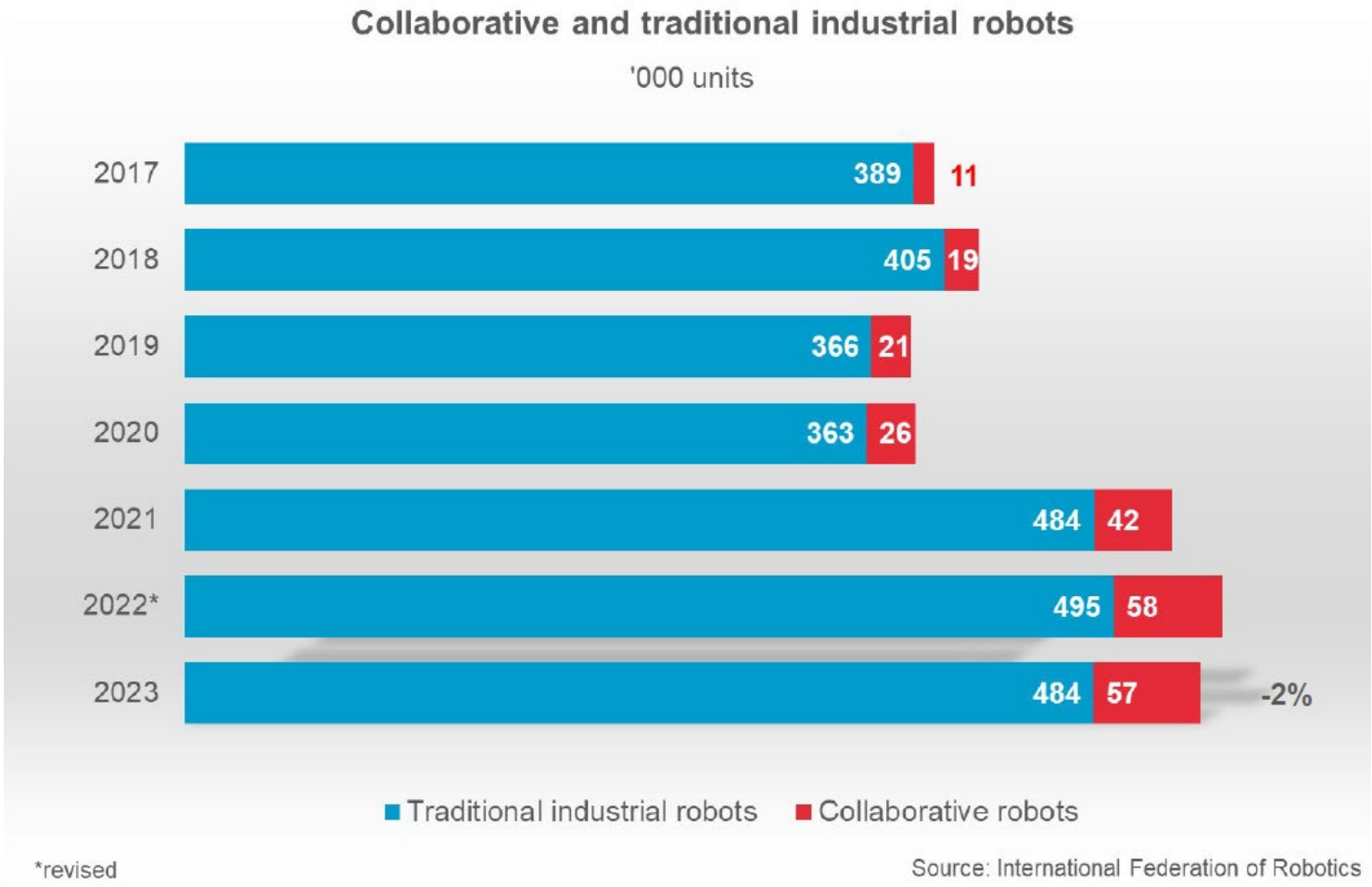
Source: A.M. Zanchettin

Installation costs of cobots

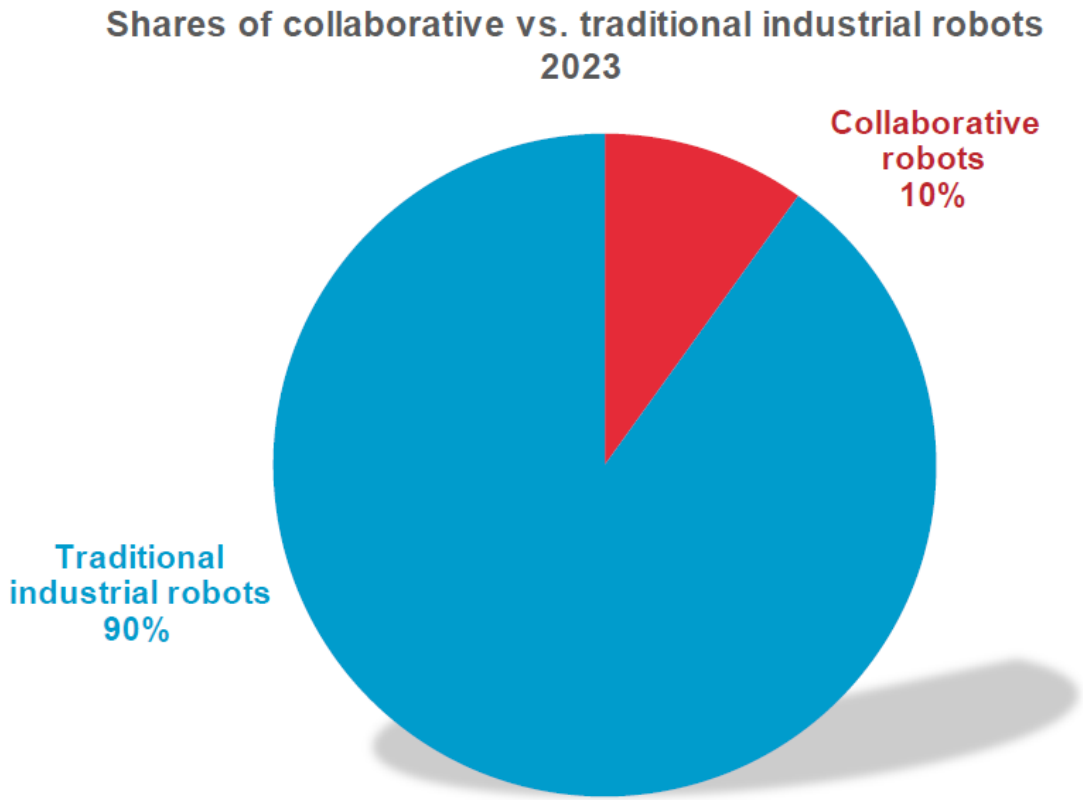
- Collaborative robots have no (or reduced) physical protection devices to allow the human operator to directly interact with them.
- The limited need of safeguarding devices allows a smaller footprint, making cobots more affordable, especially for SMEs, as compared with traditional industrial robots.



Market of the cobots

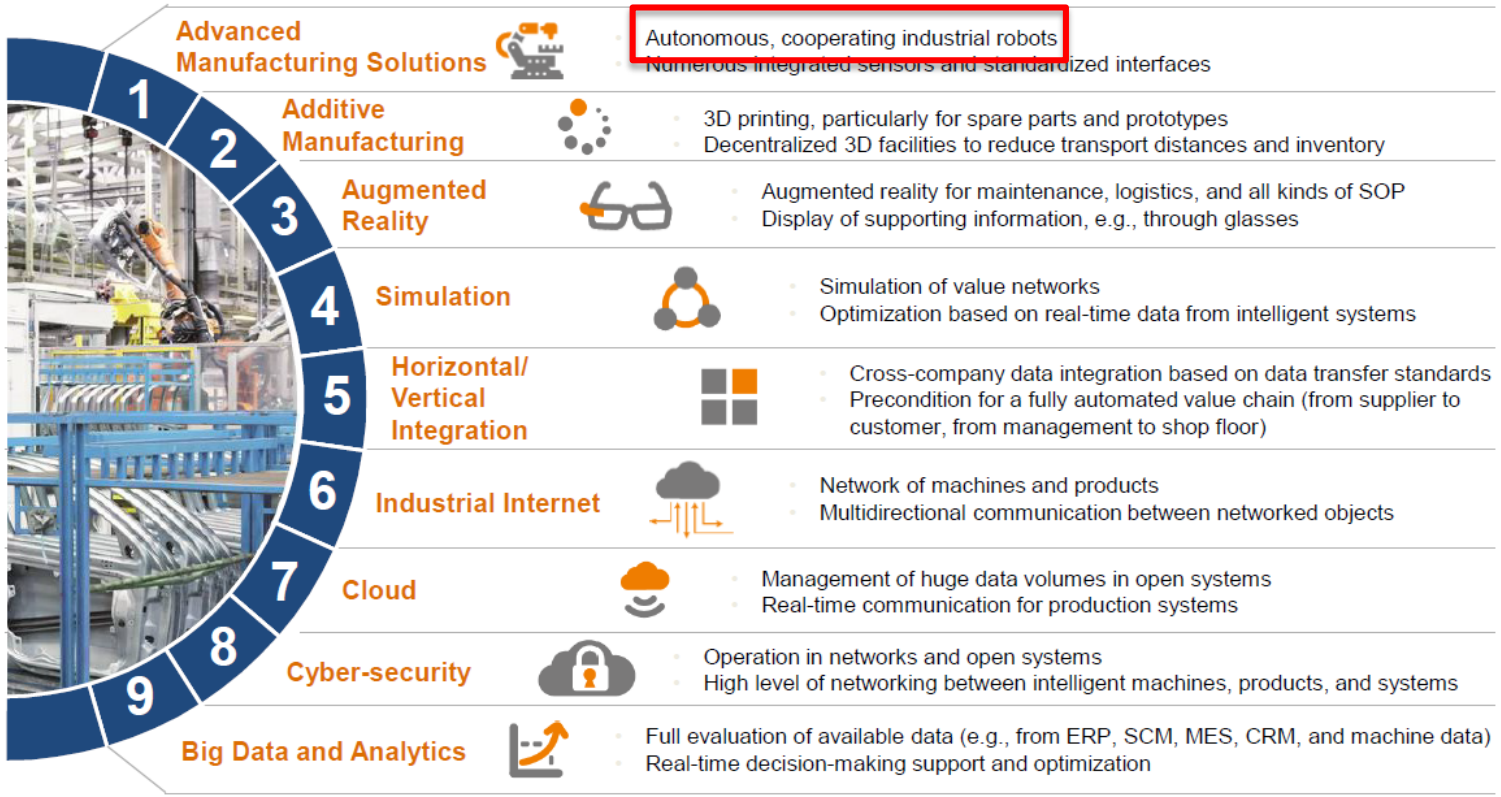


Market of the cobots



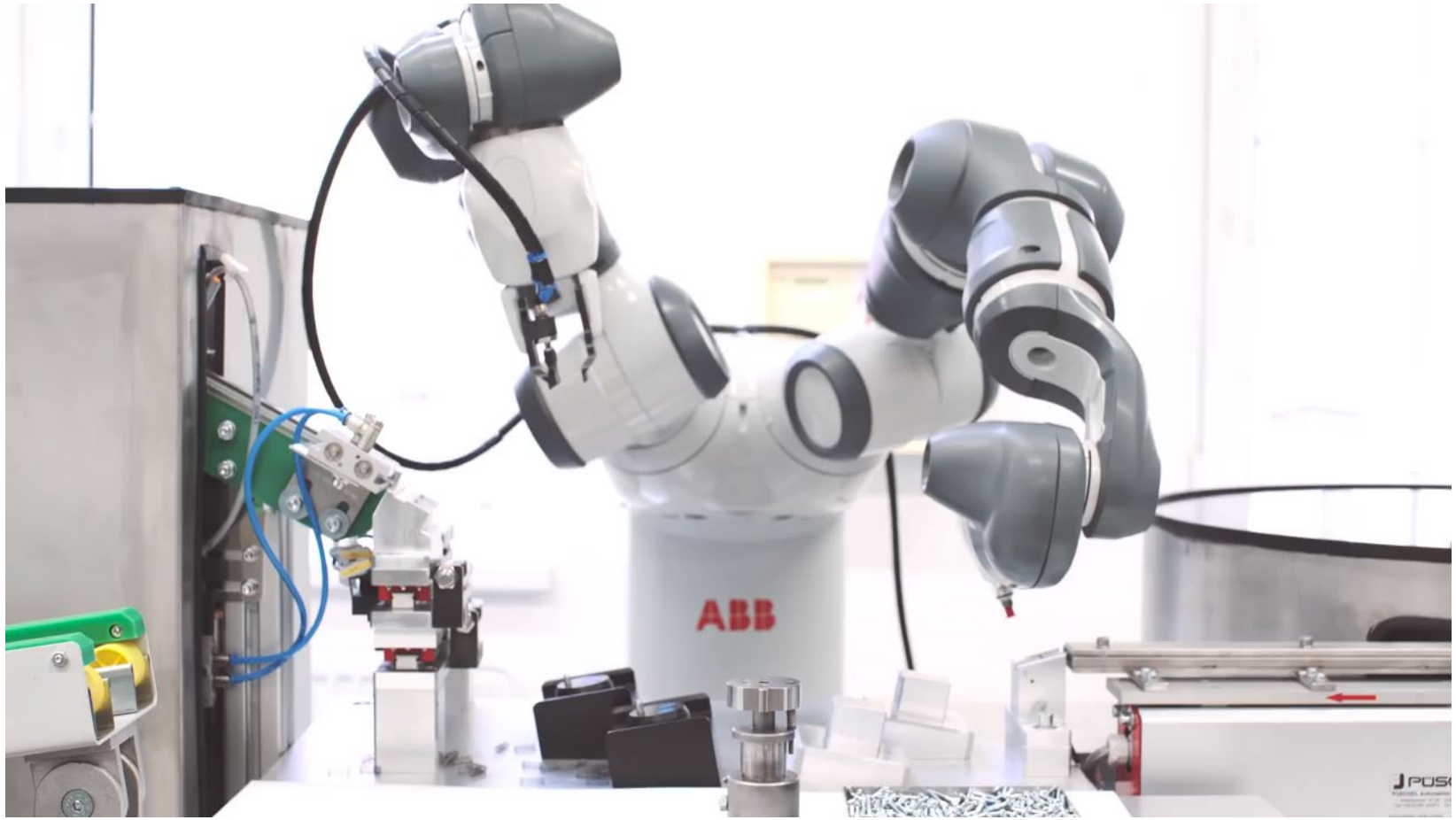
Source: International Federation of Robotics

Collaborative robotics and industry 4.0



- Collaborative robotics is considered one of the enabling technologies of the industry 4.0 paradigm
- Humans and machines are expected to actively cooperate in the smart factory

From robots to cobots



From robots to cobots



A large size cobot



Musculoskeletal disorders



Musculoskeletal disorders (MSDs) represent one of the major work-related health problems in developed countries, affecting almost 50% of industrial workers

Cost of occupational diseases in Italy

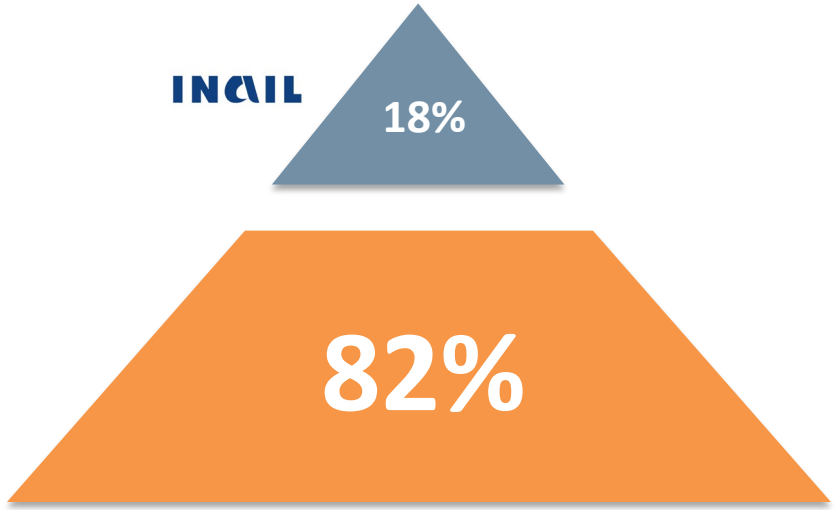




Cost (MSD):

in Italy 7.9 billions euros (0.4% GDP), source: Global Pain Index 2017

Breakdown of the cost:

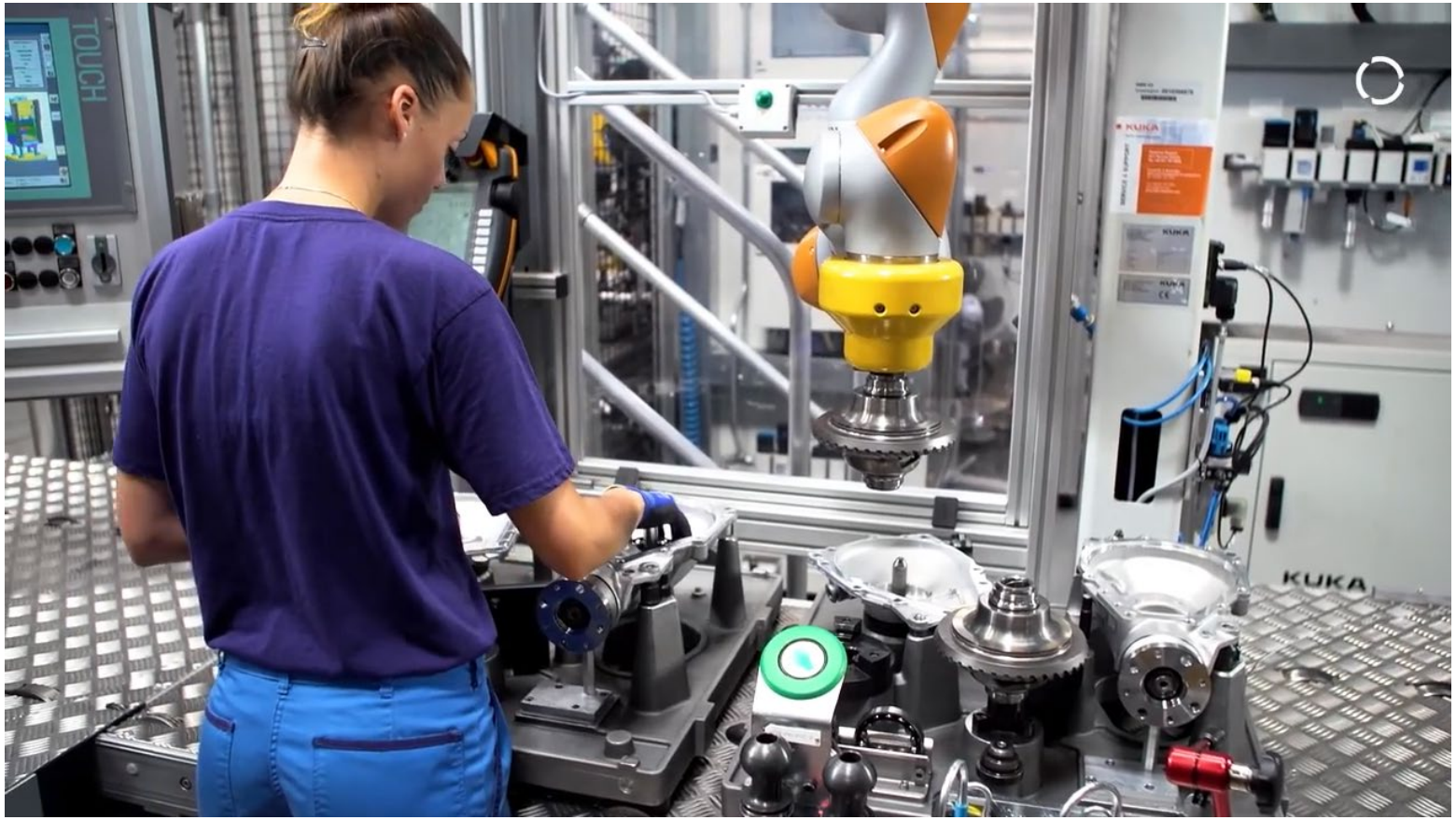


Source: OSHA

Direct costs:
compensation, health services, rehabilitation

Indirect costs:
loss of productivity, recruiting

Cobots can serve as assistants



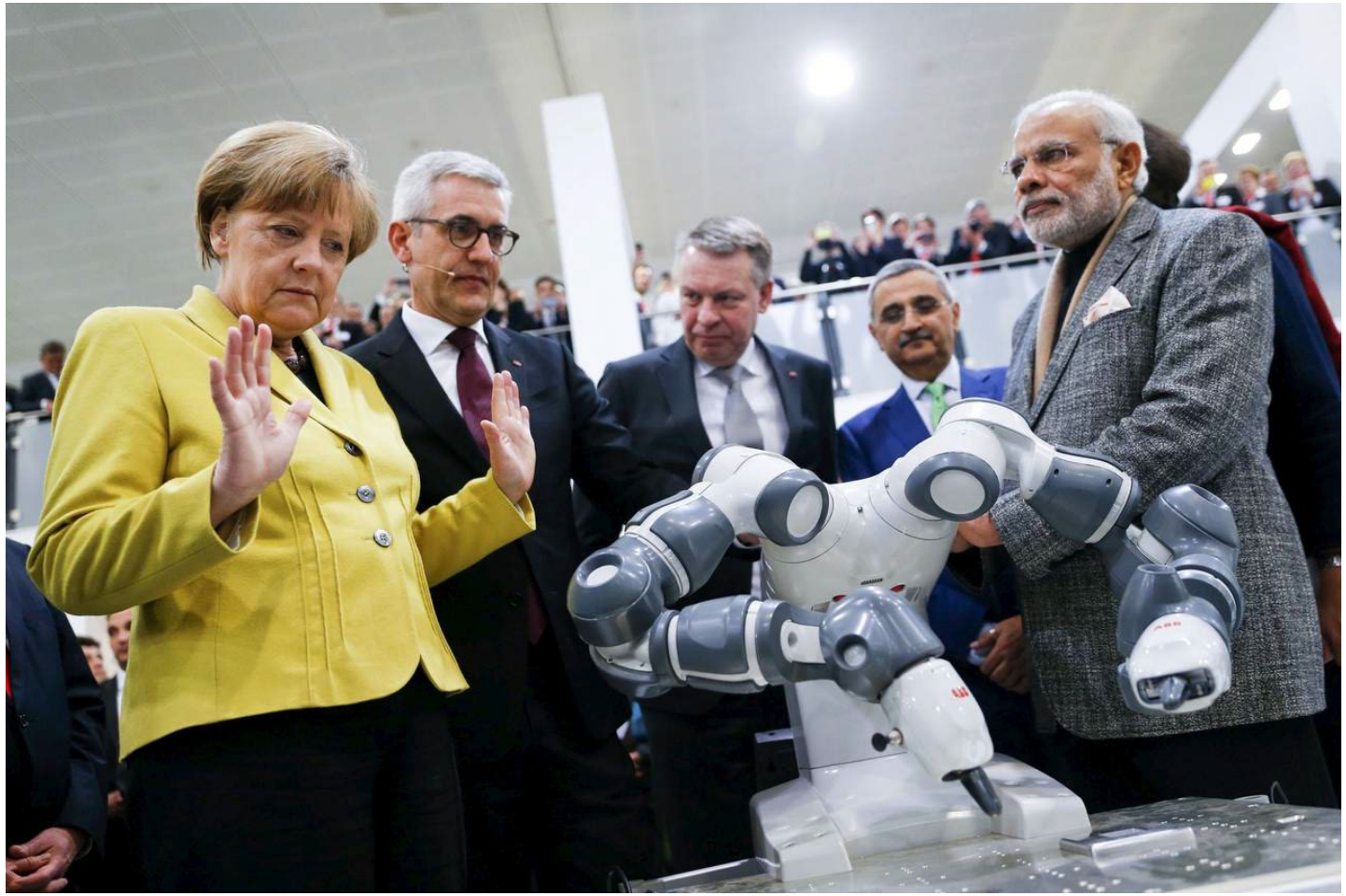
Exoskeleton can help as well



Source: COMAU



Safety



- Safety is of paramount importance when working with robots
- A robot must be able to stop in a safety certified way
- Specific safety standards exist for use of robots with humans

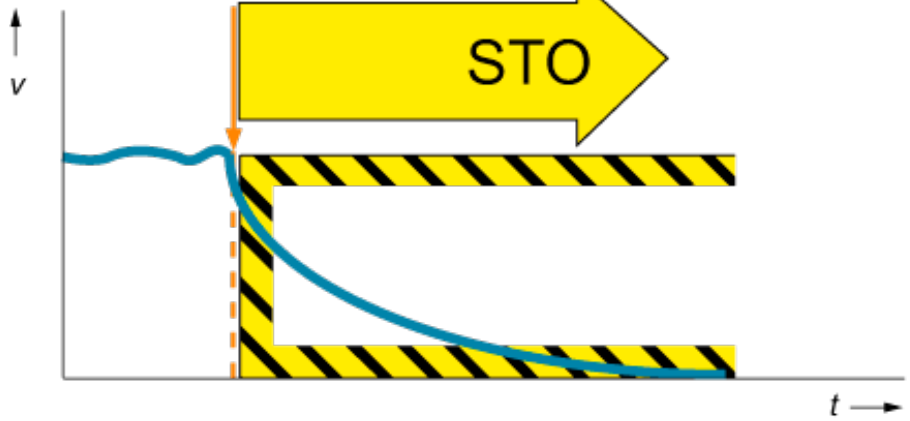
Several different braking modes

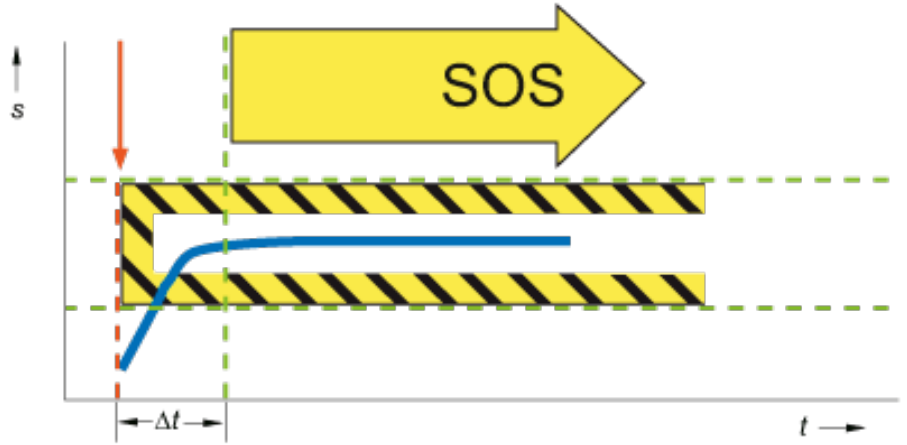
Safe torque off (STO) (category 0 stop EN 60204-1)

- The drive motor is no longer supplied with power (i.e. free of torque and force).
- The power supply to the drive is safely cut off.
- The drive cannot generate any torque, and therefore any potentially dangerous movements.

Safe operating stop (SOS)

- The drive control remains in operation (safe standstill monitoring).
- The motor can deliver the full torque to hold the current position.
- The actual position is reliably monitored.



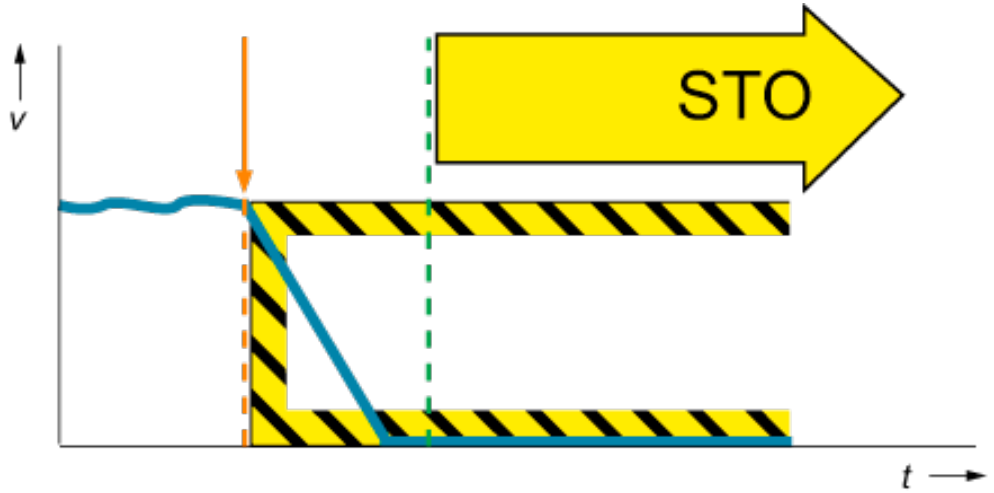


Source: A.M. Zanchettin

Several different braking modes

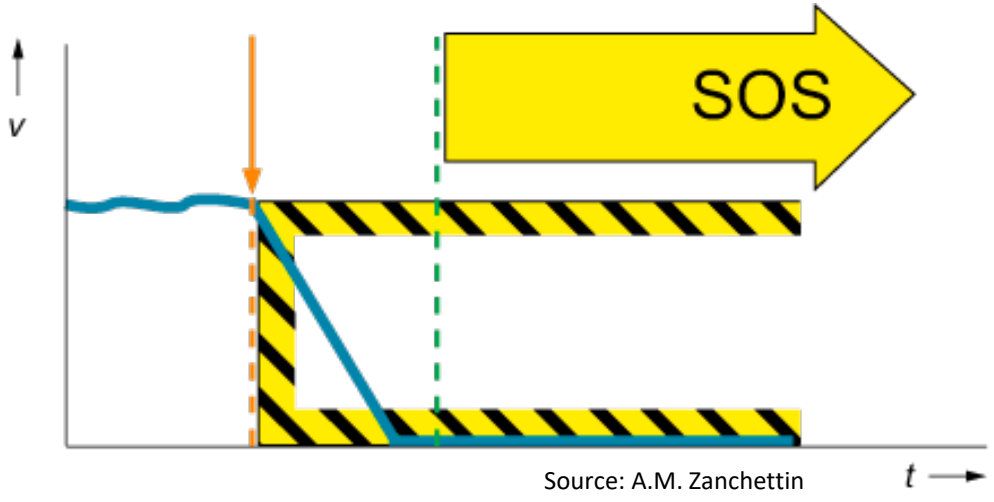
Safe stop 1 (SS1), category 1 stop (EN 60204-1)

- The drive brakes along a quick stop ramp autonomously and automatically activates the Safe Torque Off and Safe Brake Control functions (if enabled) when it reaches a complete stop.
- The deceleration phase (coasting down) is monitored through dedicated and safety-rated hardware.



Safe stop 2 (SS2), category 2 stop (EN 60204-1)

- When the SS2 function is selected, the drive brakes autonomously along a quick-stop ramp.
- Unlike SS1, the automatic speed control remains operational afterward, i.e. the motor can provide the full torque required to maintain zero speed.
- Standstill is safely monitored (Safe Operating Stop function).

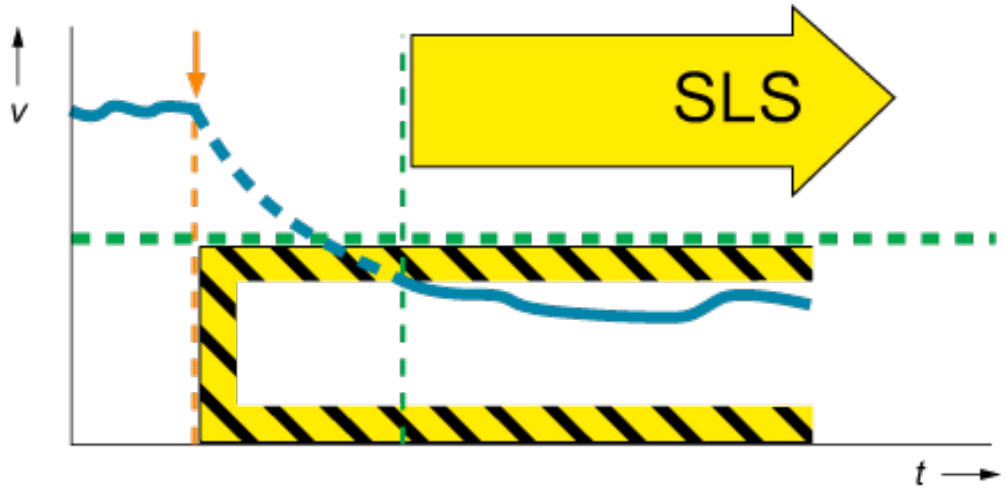


Source: A.M. Zanchettin

Safety limited speed

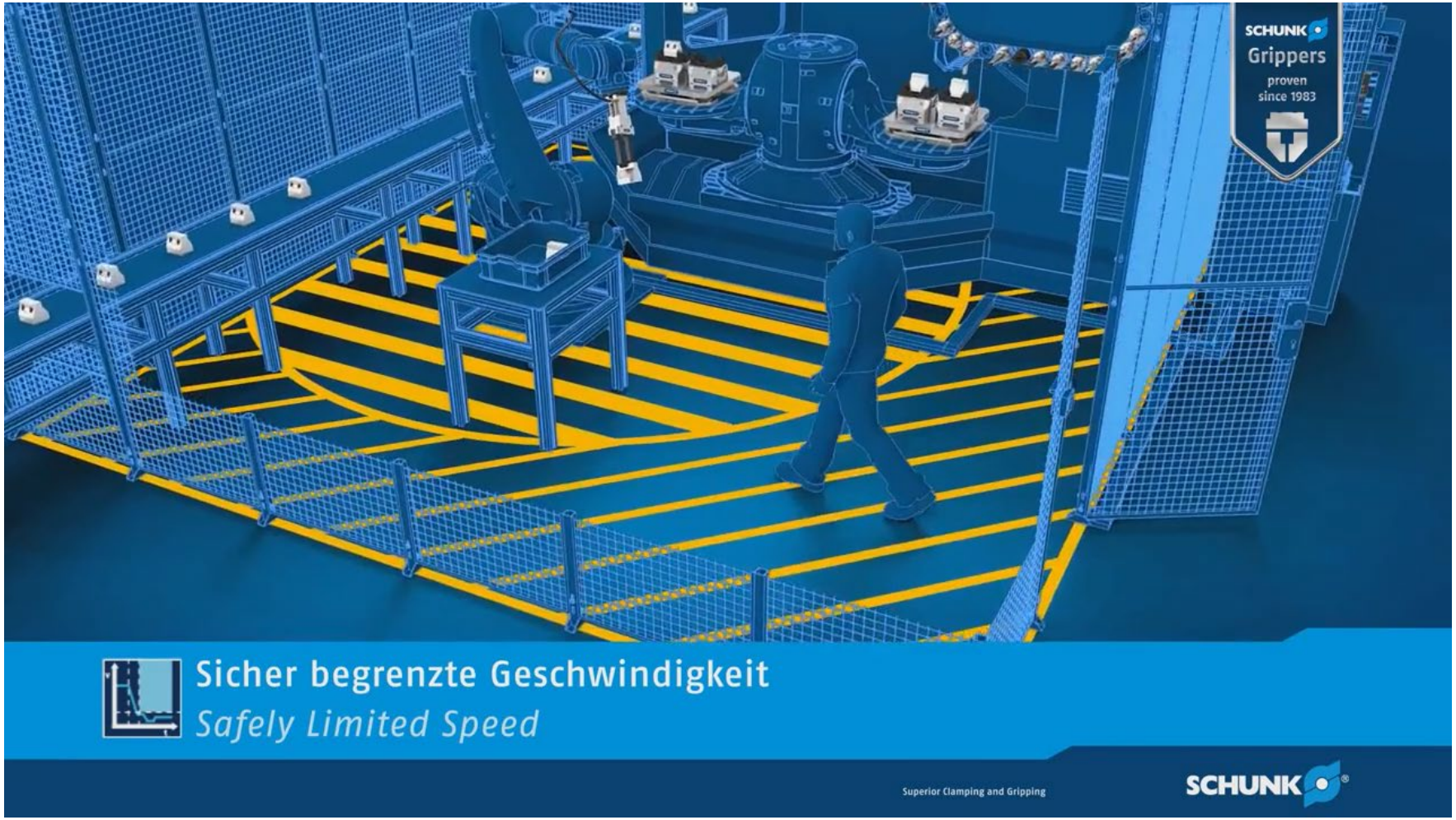
Safety limited speed (SLS):

- The drive is reliably monitored to make sure the configurable limits for speed are not exceeded.
- Depending on the application, deceleration can also be monitored until the limit is reached.
- If a violation is detected during monitoring of the limits for speed, safe pulse disabling is activated immediately

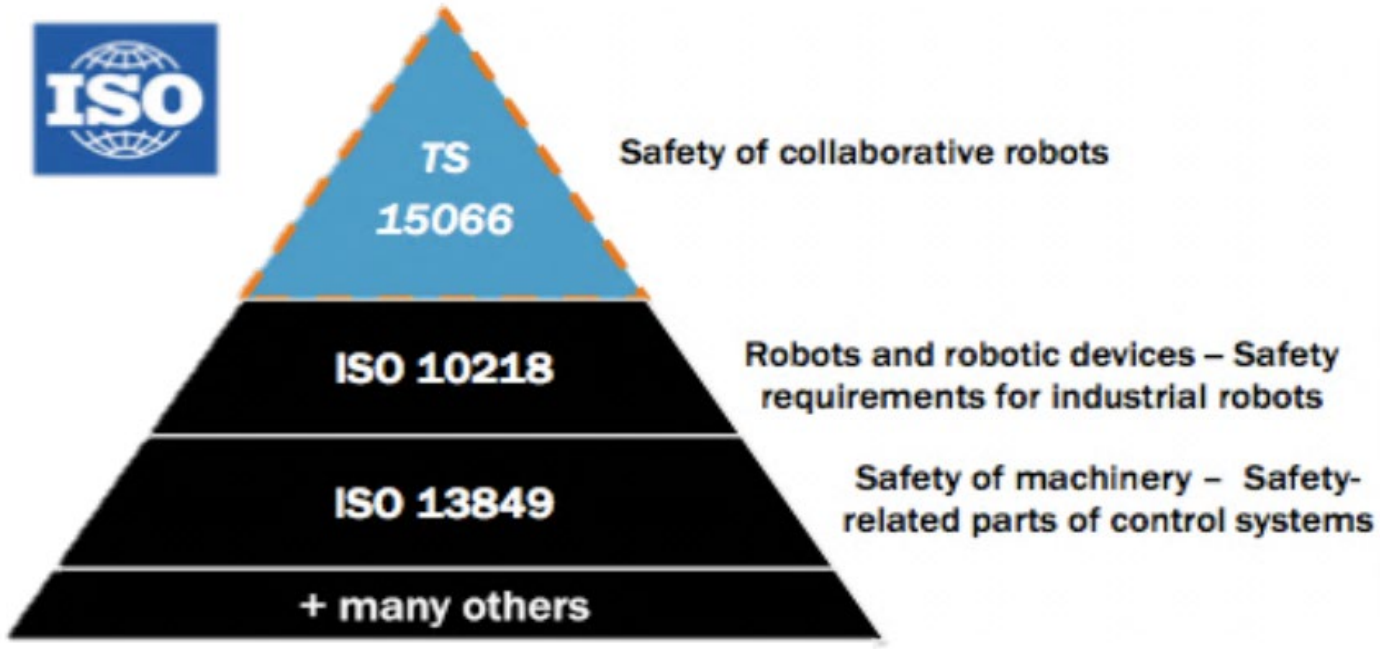


Source: A.M. Zanchettin

An integrated use of the safe functionalities

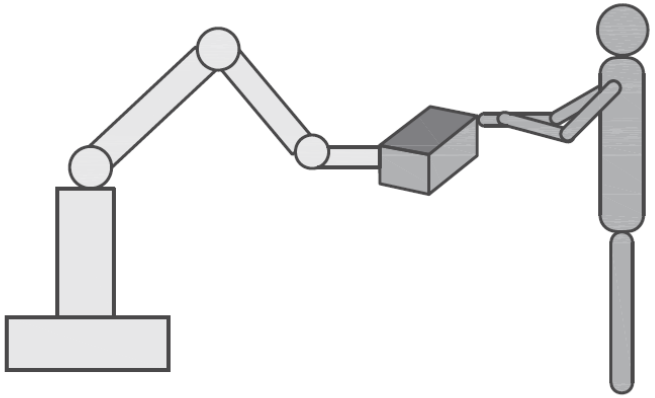


Safety standards for robotics



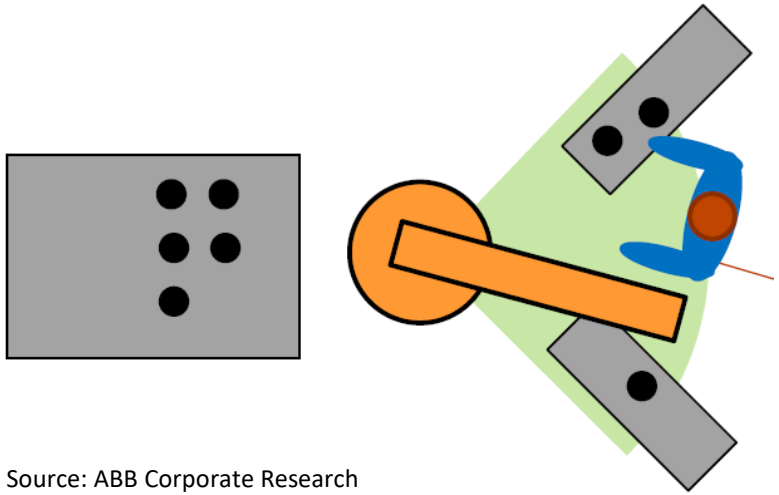
Source: A.M. Zanchettin

Safety standards: definitions



Collaborative operation

- ISO 10218-1:2011, clause 3.4
- State in which purposely designed robots work in direct cooperation with a human within a defined workspace

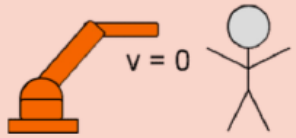
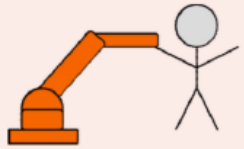
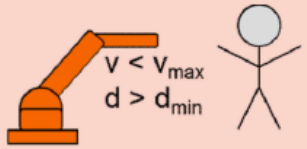
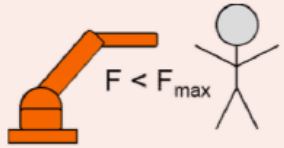


Source: ABB Corporate Research

Collaborative workspace

- ISO 10218-1:2011, clause 3.5
- Workspace within the safeguarded space where the robot and a human can perform tasks simultaneously during production operation

Types of collaborative operations

ISO 10218-1, clause	Type of collaborative operation	Main means of risk reduction	
5.10.2	Safety-rated monitored stop (Example: manual loading-station)	No robot motion when operator is in collaborative work space	
5.10.3	Hand guiding (Example: operation as assist device)	Robot motion only through direct input of operator	
5.10.4	Speed and separation monitoring (Example: replenishing parts containers)	Robot motion only when separation distance above minimum separation distance	
5.10.5	Power and force limiting by inherent design or control (Example: *ABB YuMi*® collaborative assembly robot)	In contact events, robot can only impart limited static and dynamics forces	

Source: ABB Corporate Research

Safety-rated monitored stop



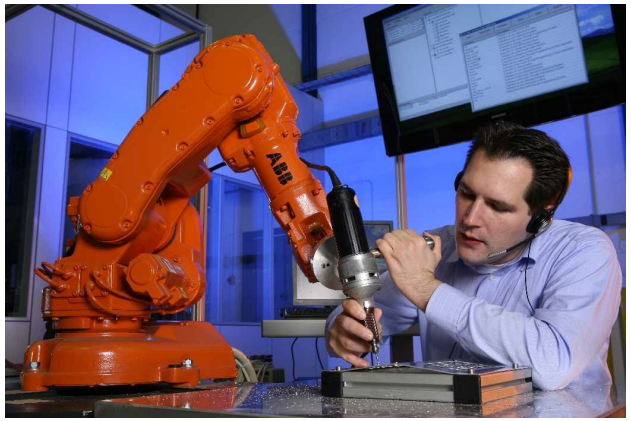
A very conservative mode. Collaboration is limited with a robot in a stand-still position. Power is still enabled and the robot is ready to resume its motion.

- **Clauses in standards and TS**
  - ISO 10218-1, clause 5.10.2
  - ISO 10218-2, clause 5.11.5.2
  - ISO/TS 15066, clause 5.5.2
- **Risk reduction**
  - Ensure robot standstill whenever a worker is in collaborative workspace
- **Achieved by**
  - Supervised standstill - Category 2 stop (IEC 60204-1)
  - Category 0 stop in case of fault (IEC 60204-1)
- **Typical applications**
  - Loading / unloading end-effector

Source: ABB Corporate Research

Hand guiding



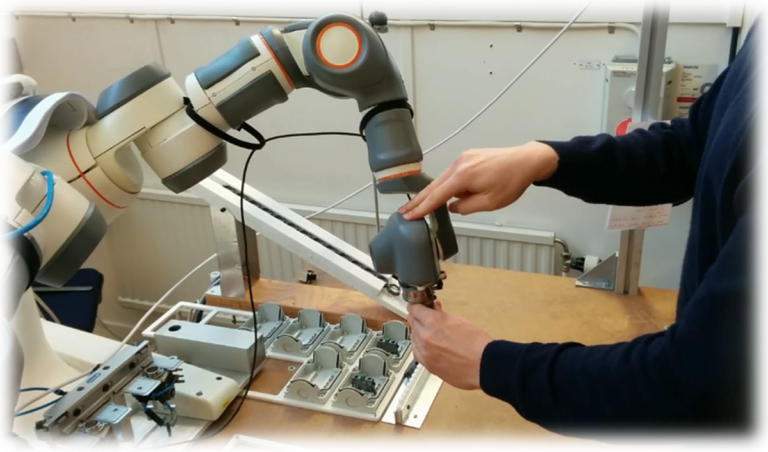


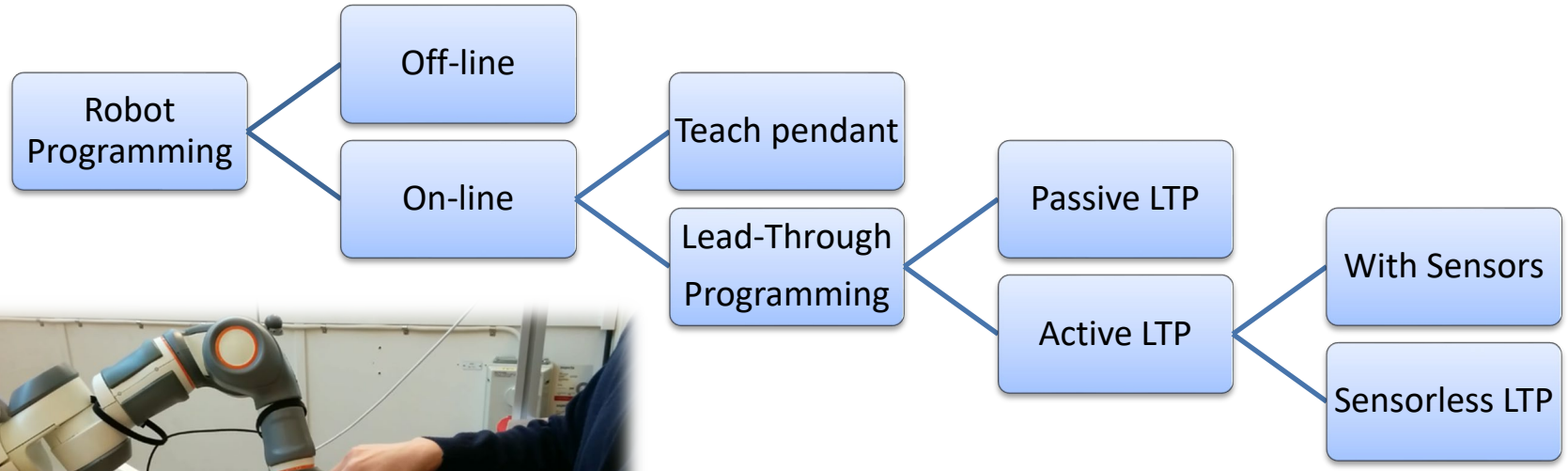
Source: ABB Corporate Research

- **Clauses in standards and TS**
  - ISO 10218-1, clause 5.10.3
  - ISO 10218-2, clause 5.11.5.3
  - ISO/TS 15066, clause 5.5.3
- **Risk reduction**
  - Provide worker with direct control over robot motion at all times in collaborative workspace
- **Achieved by**
  - Controls close to end-effector
  - Input means for motion commands
  - Emergency stop (red button)
  - Enabling device (dead man)
- **Typical applications**
  - Lift assist, load positioning

Hand guiding

Hand guiding can be used for **ergonomic purposes** (assistance in moving heavy payloads) or robot programming (**lead-through programming**)





Hand guiding





Hand guided robot:

- Potentially unsafe
- Requires expensive hardware (motors, force/torque sensor)
- Provides assistive torques (reduced inertia)
- Mid/low payload

Handling device:

- Inherently safe (passive device)
- Less expensive
- No assistive torque (no inertia reduction)
- High payload

Speed and separation monitoring



Source: ABB Corporate Research

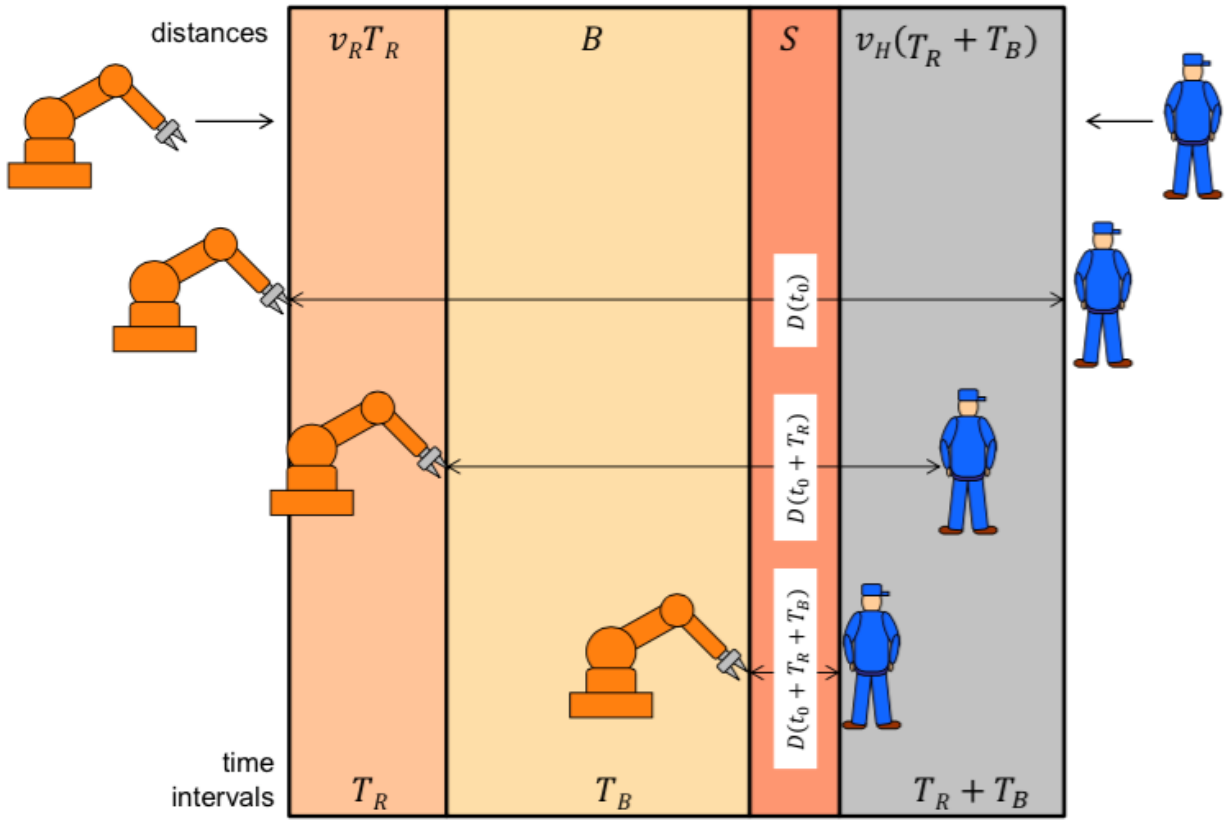
- **Clauses in standards and TS**
  - ISO 10218-1, clause 5.10.4
  - ISO 10218-2, clause 5.11.5.4
  - ISO/TS 15066, clause 5.5.4
- **Risk reduction**
  - Maintain sufficient distance between worker and robot in collaborative workspace
- **Achieved by**
  - Supervision of distance, speed
  - Protective stop if minimum separation distance or speed limit is violated
- **Typical applications**
  - Working in common area on separate tasks

Speed and separation monitoring



- The closer the operator is, the slower the robot must move
- A **safety-rated device (laser scanner)** is needed to monitor the distance of the operator from the robot

Speed and separation monitoring



v_R = robot speed

v_H = human speed

T_R = robot controller reaction time

T_B = robot stopping time

S = minimum separation distance

$D(t_0)$ = separation distance at time t_0

$$D(t_0) - v_R(T_R + T_B) - v_H(T_R + T_B) \geq S$$

The distance has to be larger than the space that the robot and the human can cover plus a minimum distance

Source: ABB Corporate Research

Speed and separation monitoring

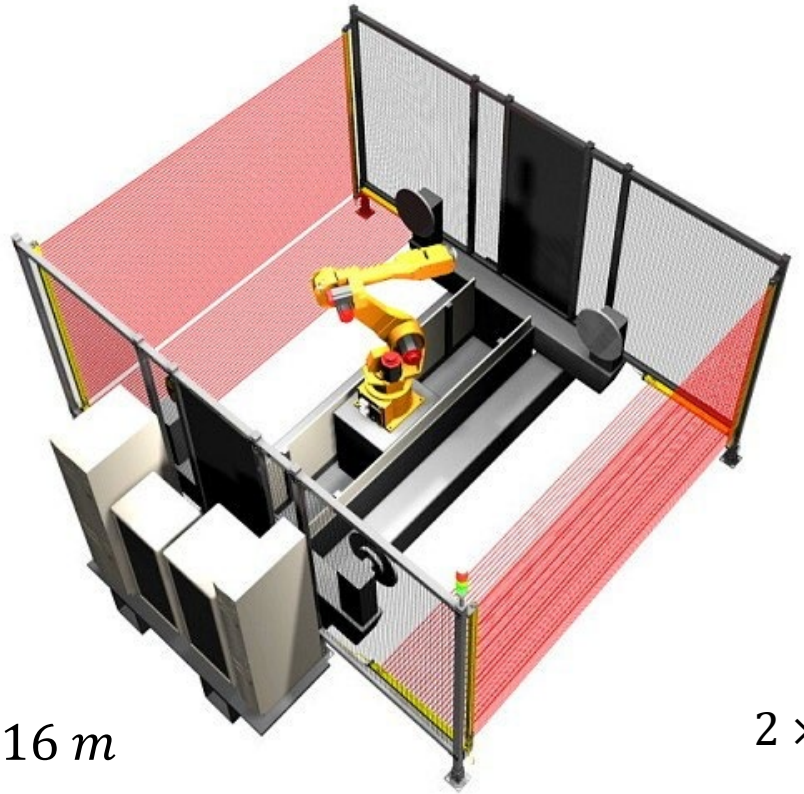
$$D(t_0) - v_R(T_R + T_B) - v_H(T_R + T_B) \geq S$$

$$v_R \leq \frac{D(t_0) - S}{T_R + T_B} - v_H$$

- If the **human speed** is not monitored, it can be assumed to be $v_H = 1.6\ m/s$ in the direction that reduces the separation distance the most
- The **robot speed** v_R can be either constant or variable
  - if the robot's speed v_R is not being monitored, the system design shall assume that v_R is the maximum speed of the robot
  - if the robot's speed is being monitored, the system design may use the current speed of the robot

Speed and separation monitoring: a conservative design

- v_H is in the order of 1.6 m/s (walking speed)
- T_R in the order of 10 ms (reaction time of the robot controller)
- T_B in the order of 1 s (braking time of the robot actuators)



$$D(t_0) \geq v_H(T_R + T_B) = 1.6 \times 1.01 = 1.616\, m$$

Distance to be kept for whatever robot speed

Assuming the reach of the robot to be of 1 m (3.14 m^2), the cell must be large at least

$$2 \times (1 + 1.616) = 5.232\, m\ (27\, m^2)$$

(almost 10x the workspace of the robot)

Speed and separation monitoring



https://www.youtube.com/watch?v=Z7DgzHC9e9E

Power and force limiting



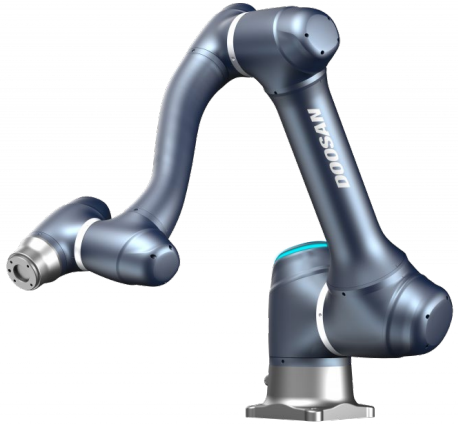
Source: ABB



Source: KUKA



Source: Universal Robots



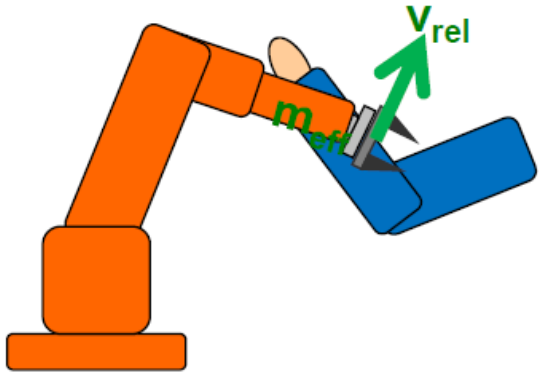
Source: Doosan

- **Clauses in standards and TS**
  - ISO 10218-1, clause 5.10.5
  - ISO 10218-2, clause 5.11.5.5
  - ISO/TS 15066, clause 5.5.5
- **Risk reduction**
  - Limiting mechanical loading of human-body parts by moving parts of robot, end-effector or work piece
- **Achieved by**
  - Low inertia, suitable geometry and material, sensory input, control functions, …
- **Typical applications**
  - Mixed environment, involving possibility of transient and/or quasi-static physical contact

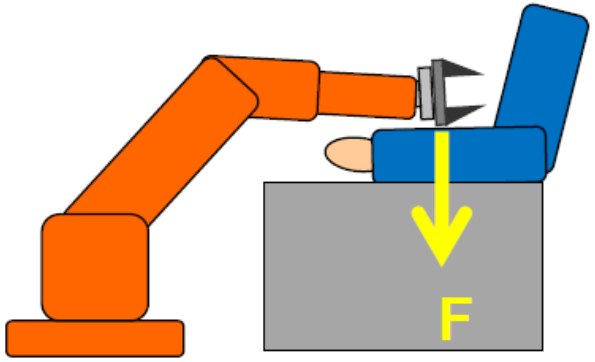
Source: ABB Corporate Research

Transient and quasi-static contacts

Transient contact

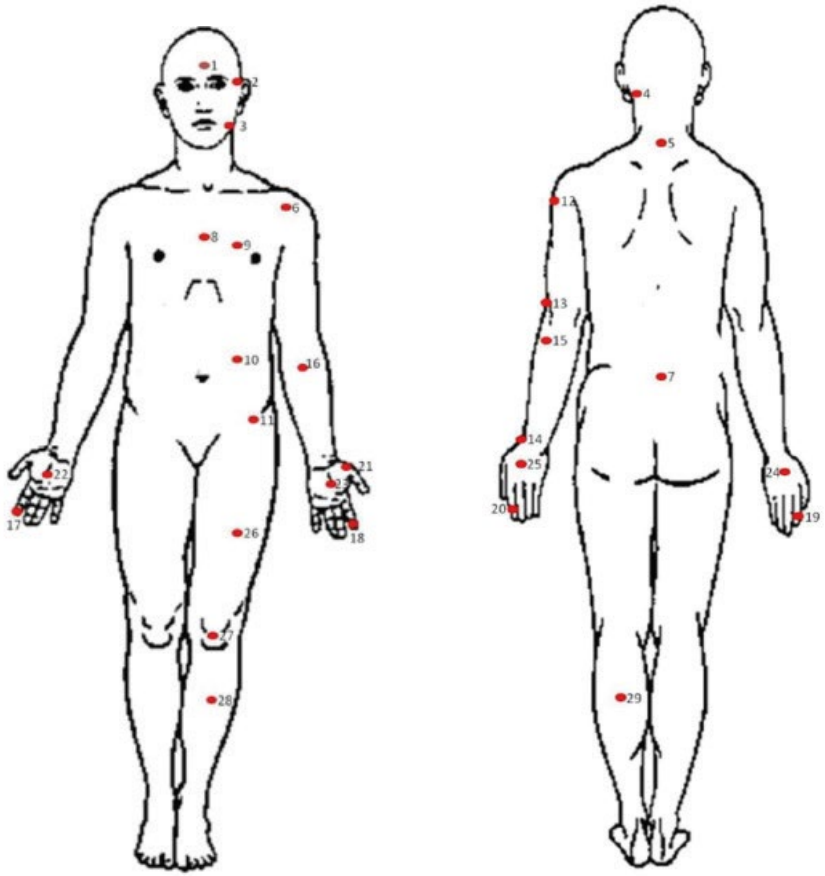


Quasi-static contact



Source: ABB Corporate Research

Biomechanical limits



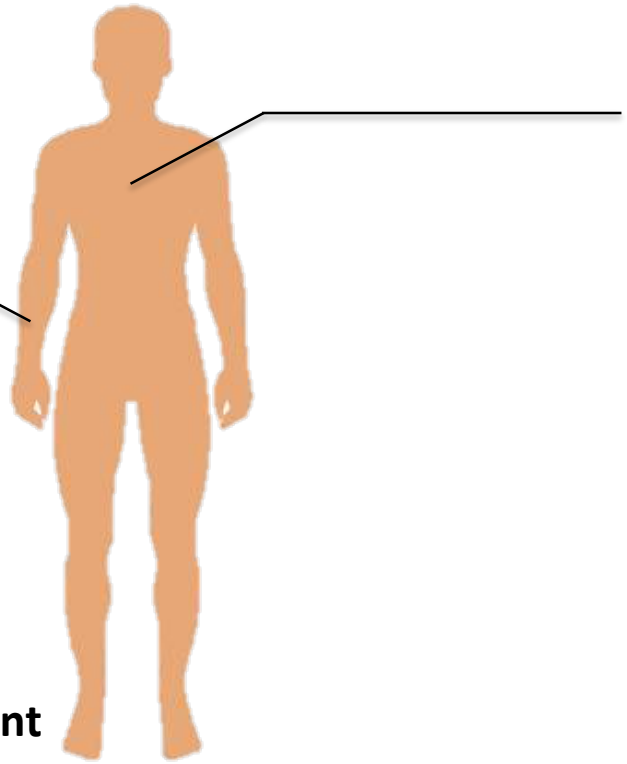
ISO/TS 15066, Annex A

- Establishes threshold limit values on the collaborative robot system, particularly on power and force limiting applications.
- Based on pain sensitivity thresholds
- Can be used to establish pressure and force limit values for various body areas using a body model.

Biomechanical limits: a few examples

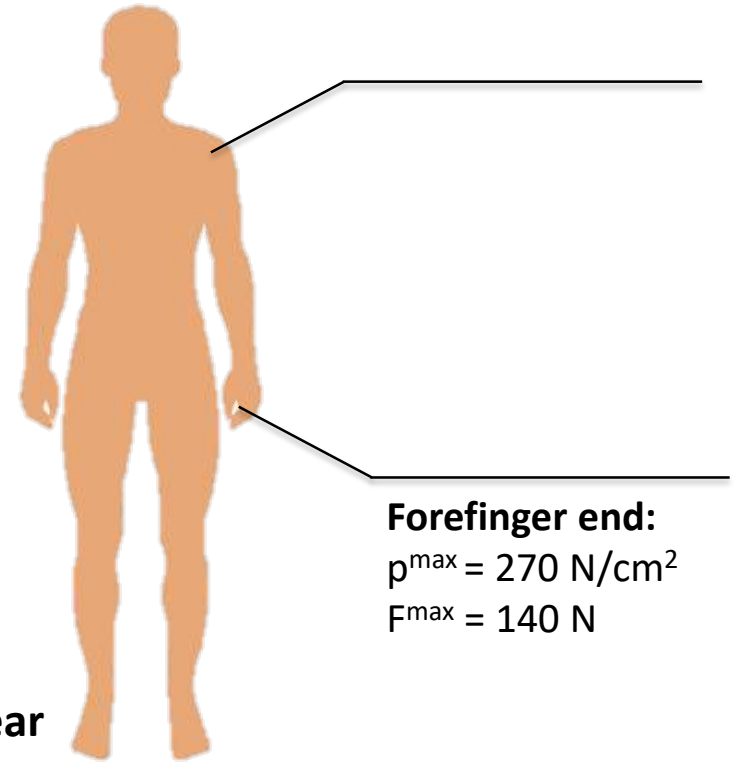
Sternum:
$p^{max} = 120\ N/cm^2$
$F^{max} = 140\ N$

Arm nerve:
$p^{max} = 180\ N/cm^2$
$F^{max} = 160\ N$



Front

Deltoid:
$p^{max} = 190\ N/cm^2$
$F^{max} = 150\ N$

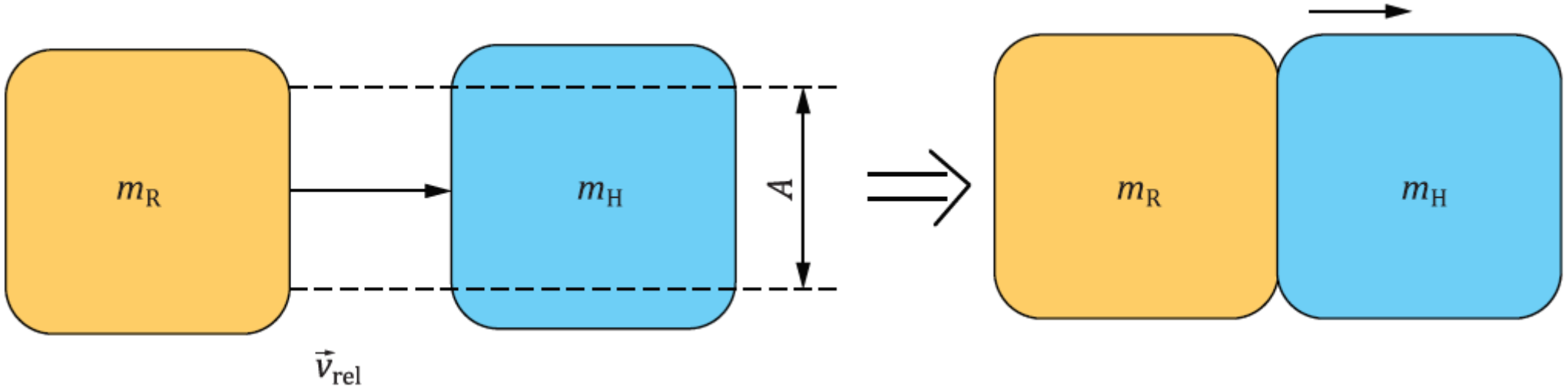


Rear

Forefinger end:
$p^{max} = 270\ N/cm^2$
$F^{max} = 140\ N$

Modelling contacts

In order to compute the speed limits, **we need a model of the contact**



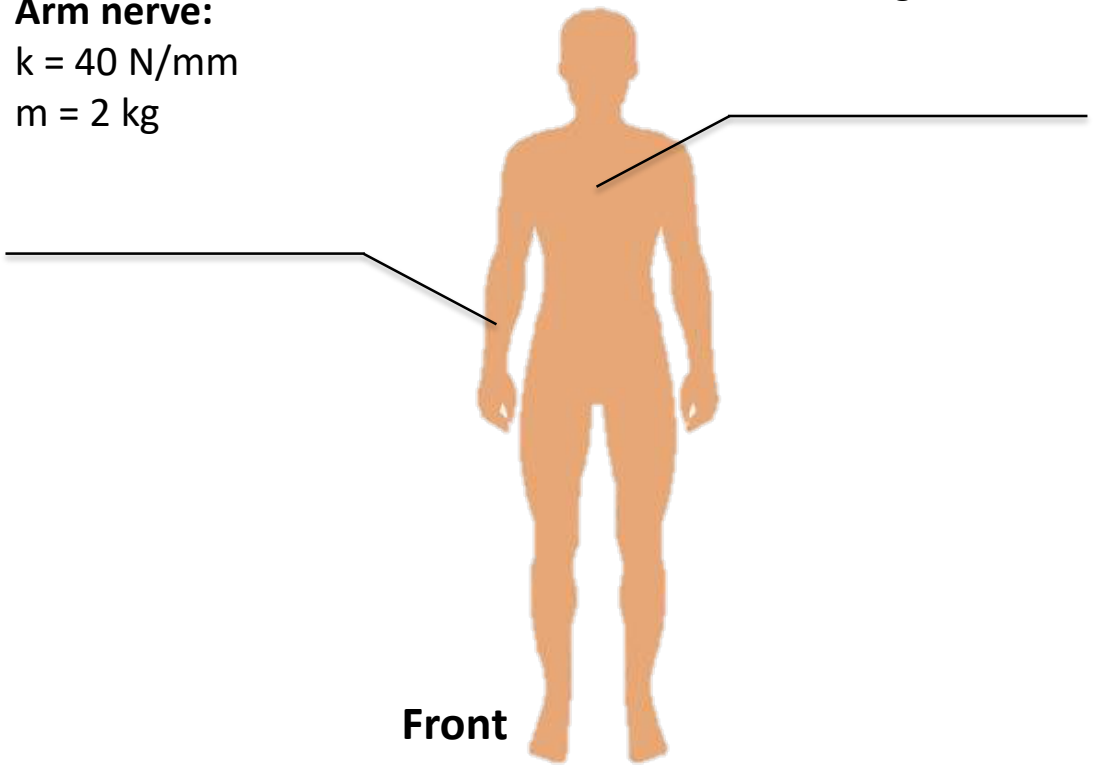
A	area of contact between robot and body region
m_H	effective mass of human body region
m_R	effective mass of robot as a function of robot posture and motion
v_{rel}	relative speed between robot and human body region

A **fully inelastic contact situation** is assumed, which corresponds to a worst case assumption.
The relative kinetic energy is assumed to be fully deposited in the affected body region.

Effective masses and spring constants of the human body parts: a few examples

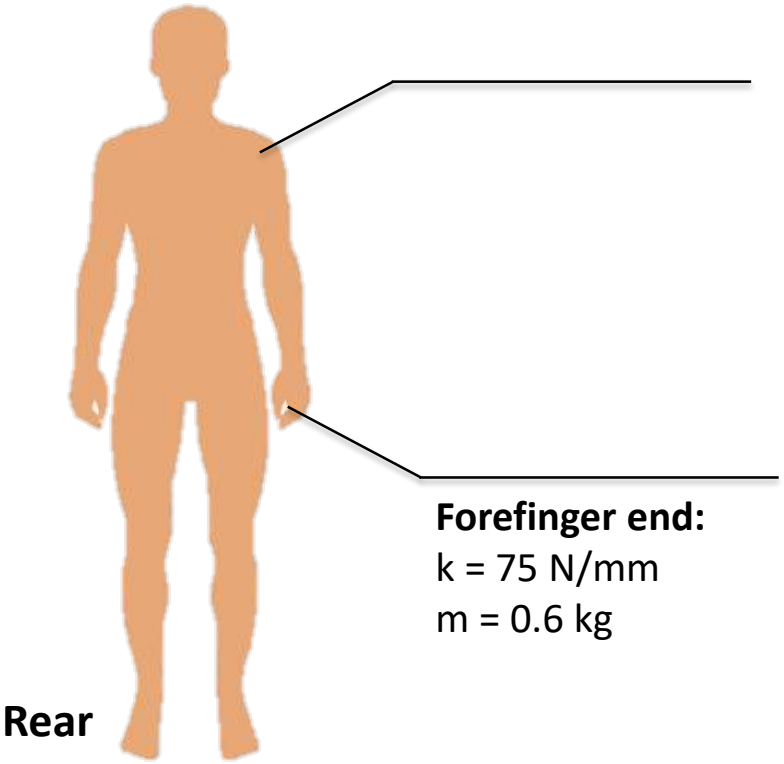
Sternum:
k = 25 N/mm
m = 40 kg

Arm nerve:
k = 40 N/mm
m = 2 kg



Front

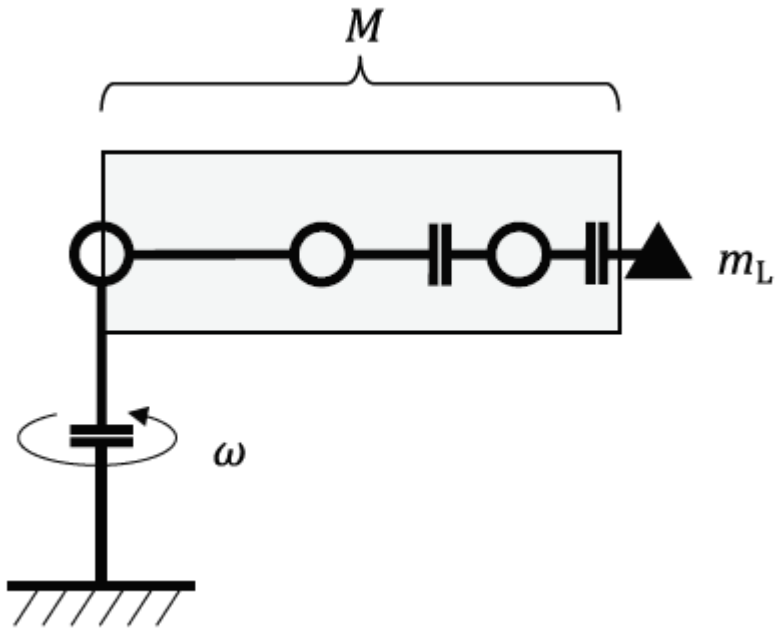
Deltoid:
k = 35 N/mm
m = 40 kg



Forefinger end:
k = 75 N/mm
m = 0.6 kg

Rear

Effective mass of the robot



Source: ISO

M total mass of the moving parts of the robot

m_L effective payload

Effective mass of the robot:

$$m_R = \frac{M}{2} + m_L$$

Speed limits

We equate the spring energy of the human body region to the total kinetic energy transferred during the impact, assuming fully inelastic contact

$$\frac{1}{2}\frac{F^2}{k} = \frac{1}{2} m_{red} v_{rel}^2$$

where m_{red} is the reduced mass:

$$m_{red} = \left(\frac{1}{m_H} + \frac{1}{m_R} \right)^{-1}$$

(can be proven studying the transfer of kinetic energy)

The speed limit can then be obtained as:

$$v_{rel_{max}} = \frac{F_{max}}{\sqrt{k m_{red}}} = \frac{p_{max} A}{\sqrt{k m_{red}}}$$

The contact area A is the smallest of the surface areas of the robot or the operator.

Speed limits

$$v_{rel_{max}} = \frac{F_{max}}{\sqrt{km_{red}}} = \frac{p_{max}A}{\sqrt{km_{red}}}$$

- the higher the area A, the larger the limit: this is way new generation robots have smooth surfaces and no pinch points
- the lighter the robot, the larger the limit: this is way cobots are typically light-weight with reduced payload

Clamping (quasi-static) case

In the clamping case, the mass of the human can be considered as of infinite value.
Therefore the following formula can be used for the speed limit:

$$v_{rel_{max}} = \frac{F_{max}}{\sqrt{k m_R}} = \frac{p_{max} A}{\sqrt{k m_R}}$$

Collaborative grippers



Source: ABB

- It is not just the robot that has to be collaborative
- The **grippers** as well need to be designed for safe human-robot collaboration

Robot as an element in the collaborative station



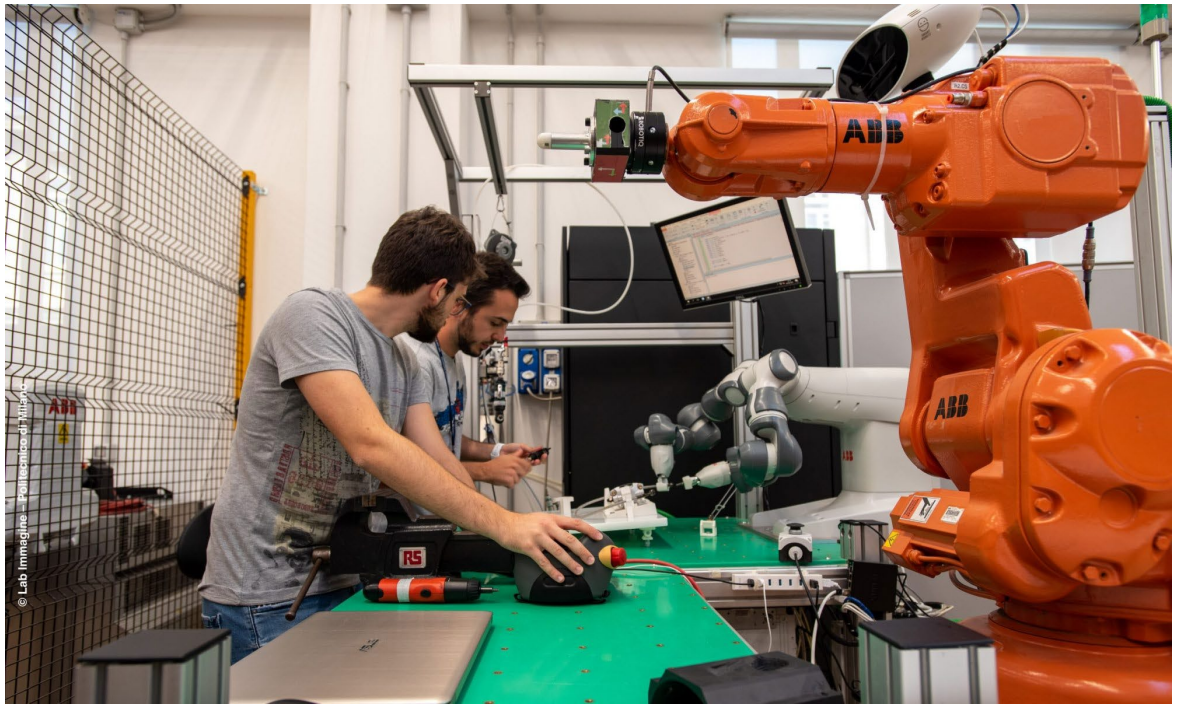
The robot is simply a component in a final collaborative robot system and is not in itself sufficient for a safe collaborative operation. The collaborative operation applications are dynamic and shall be determined by the **risk assessment** performed during the application system design.

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ME chatronics and R obotics L aboratory for IN novation

Robotic available facilities

Robotic manipulators
Dual arm robot
Robot controllers
Haptic devices
Cameras



People

3 faculty members
7÷8 PhD students and research grants assignees
40+ MSc students/year

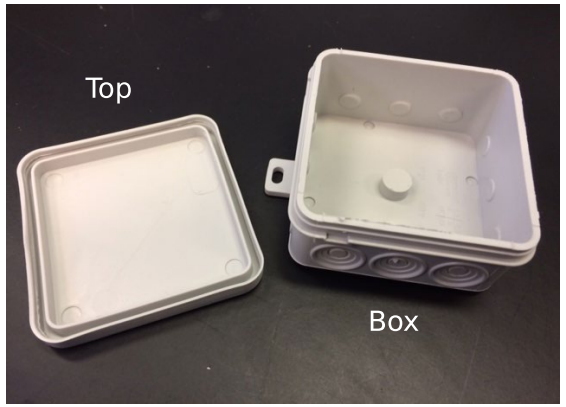
merlin.deib.polimi.it

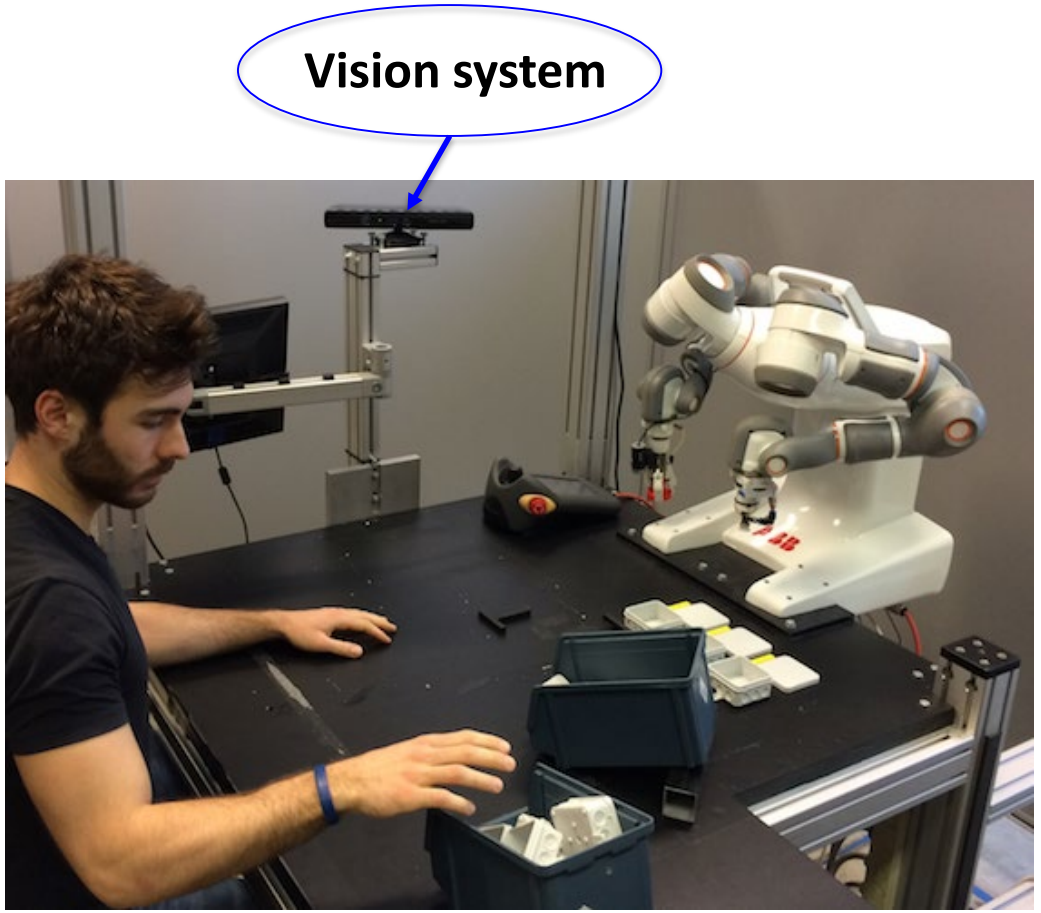
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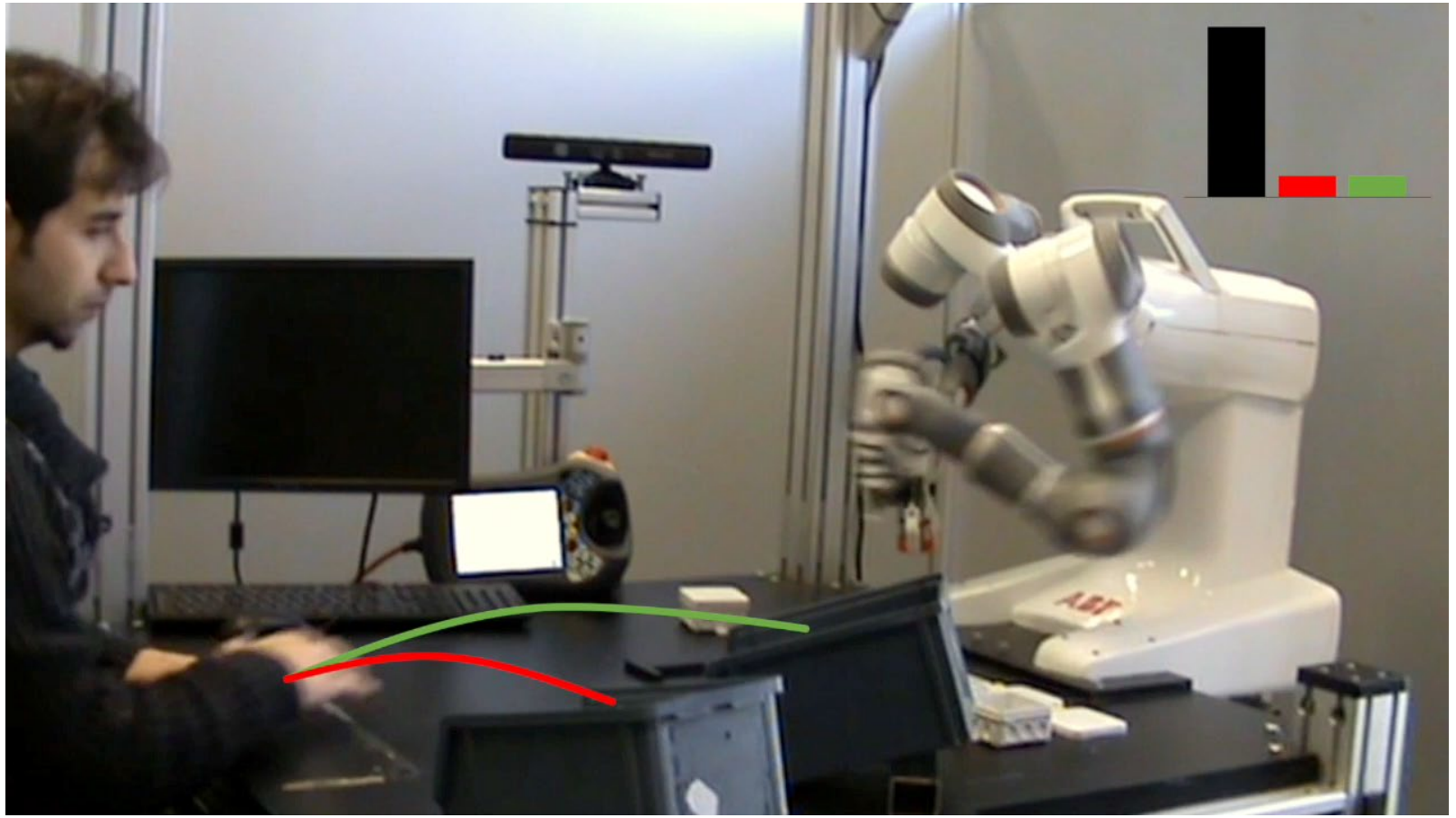
Cobots as intelligent fellow co-workers

Through a vision system the robot is able to infer the intention of the human operator and decide the most convenient action to perform according to the collaborative task.





Collaborative operation



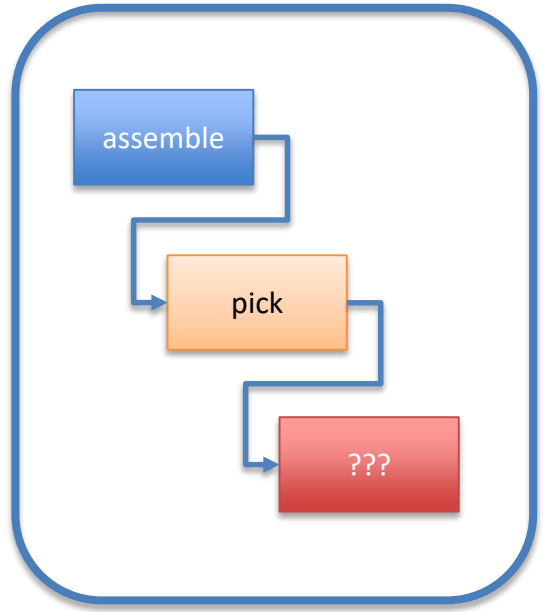
Link to the video: https://www.youtube.com/watch?v=P1p1-hejjaQ

Feedback to the operator



Prediction of activity patterns



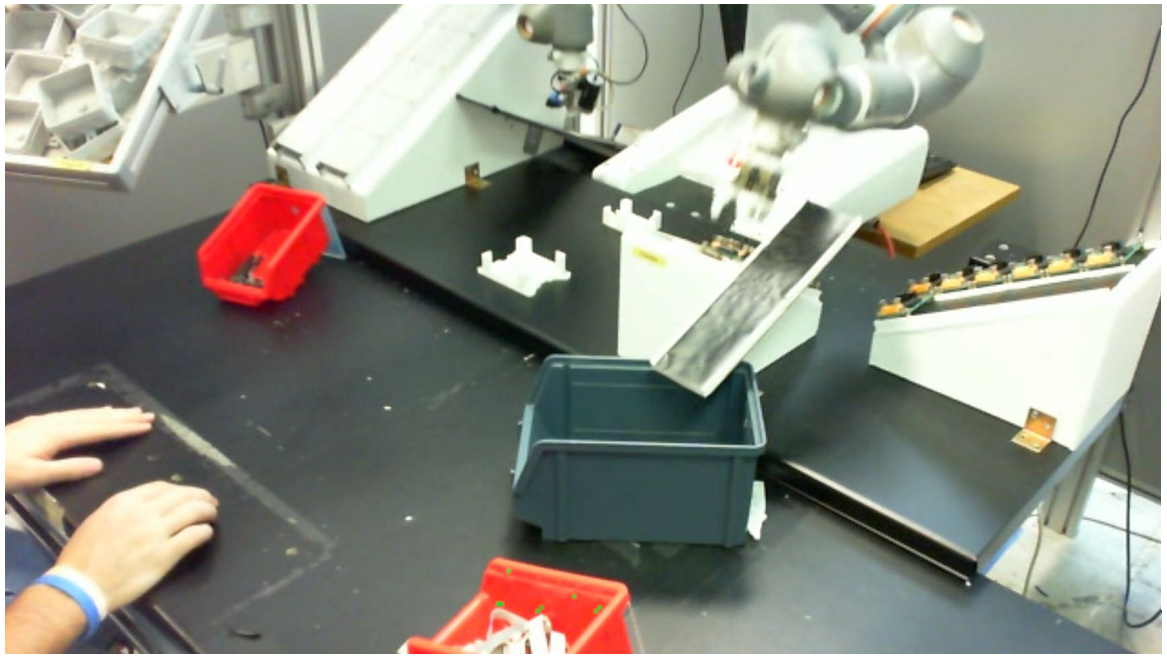


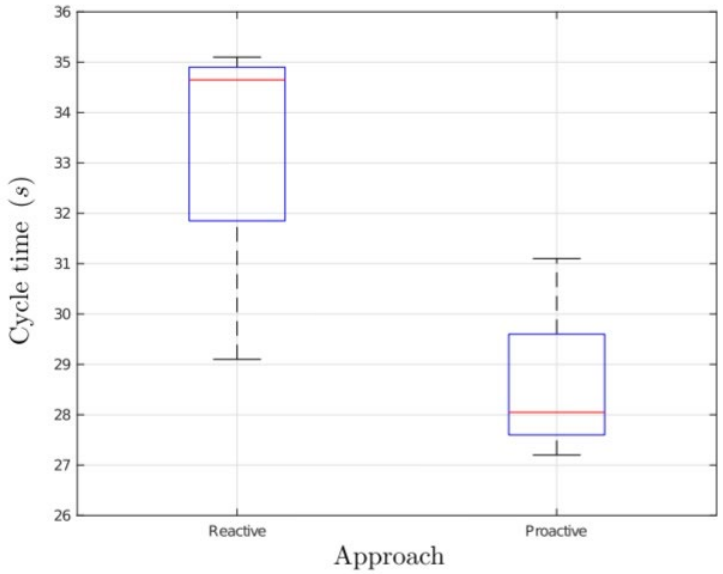
Prediction of activity patterns

Link to the video: https://www.youtube.com/watch?v=yHK2vXf0mrk

The robot perceives the human and predicts when assistance is needed. In the meanwhile, it keeps doing its job. As a result, the robot is always ready to help, and productivity is maximized.

Prediction of activity patterns





Cycle time is reduced by 20%

Integration with mix-reality



Intelligent collaborative robotics

